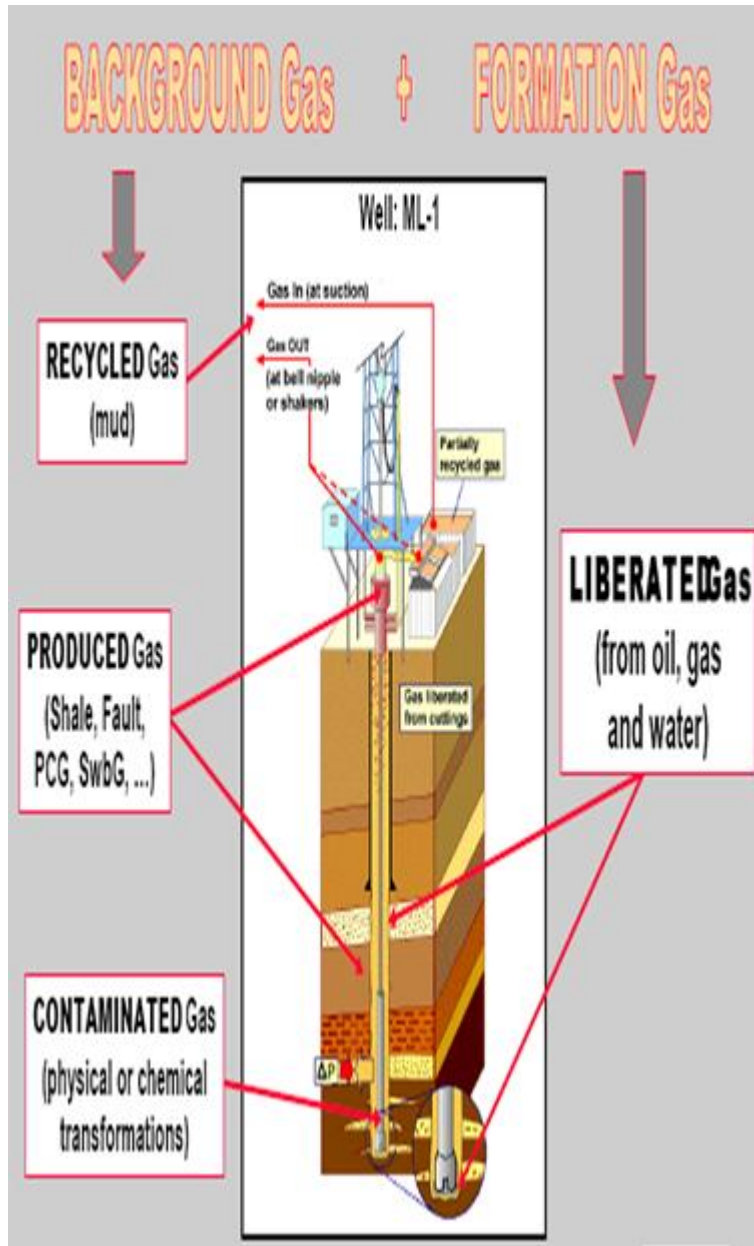


GAS DATA, QC & INTERPRETATION

Mud Logging Technical Training

PT. GEO OPTI OILFIELD

**GAS DATA, QC
& INTERPRETATION**

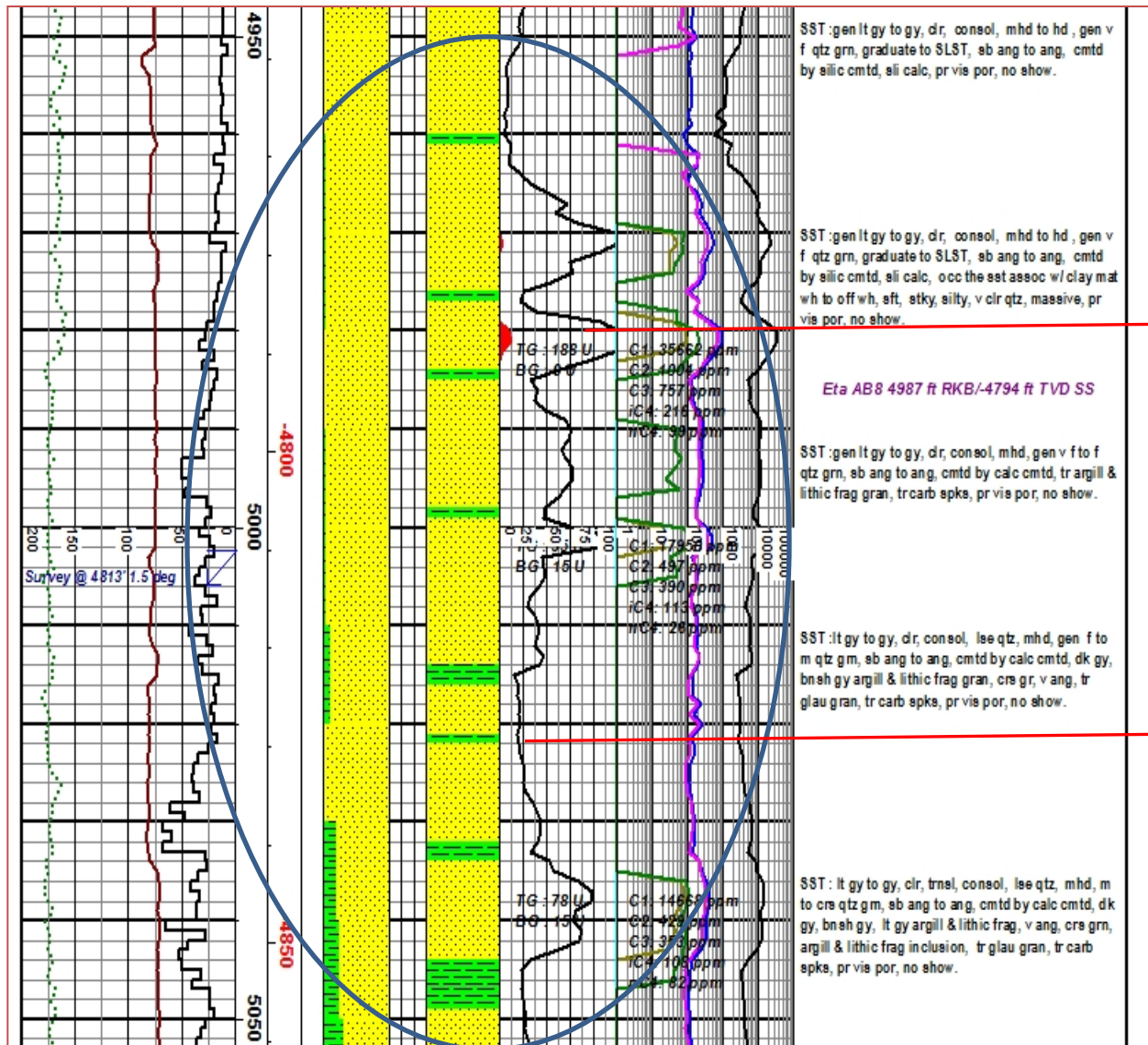


Banyak bukti yang menunjukkan bahwanya kesalahan tafsir pada data deteksi gas. Pertanyaan itu sering diajukan, "Seberapa besar gas yang bisa di diharapkan dari zona yang pada sumur pemboran?" Kesalahpahaman seperti itu seringkali dilacak karena kurangnya pemahaman terhadap prinsip dasar pendeteksian dan interpretasi gas.

Untuk mengilustrasikan prinsip-prinsip dasar ini, model pengeboran yang menunjukkan efek dari penetrasi bit. Model ini dianalisis untuk menjelaskan respons deteksi gas teoretis terhadap penetrasi zona hidrokarbon. Karakteristik acara gas, sebagaimana ditransmisikan ke permukaan oleh fluida pengeboran, secara khusus terkait dengan penetrasi bit. Analisis yang cermat untuk model pengeboran berasal empat

Classifications of gas present in the drilling fluid. These are:

1. Liberated gas
2. Produced gas
3. Recycled gas
4. Contamination gas

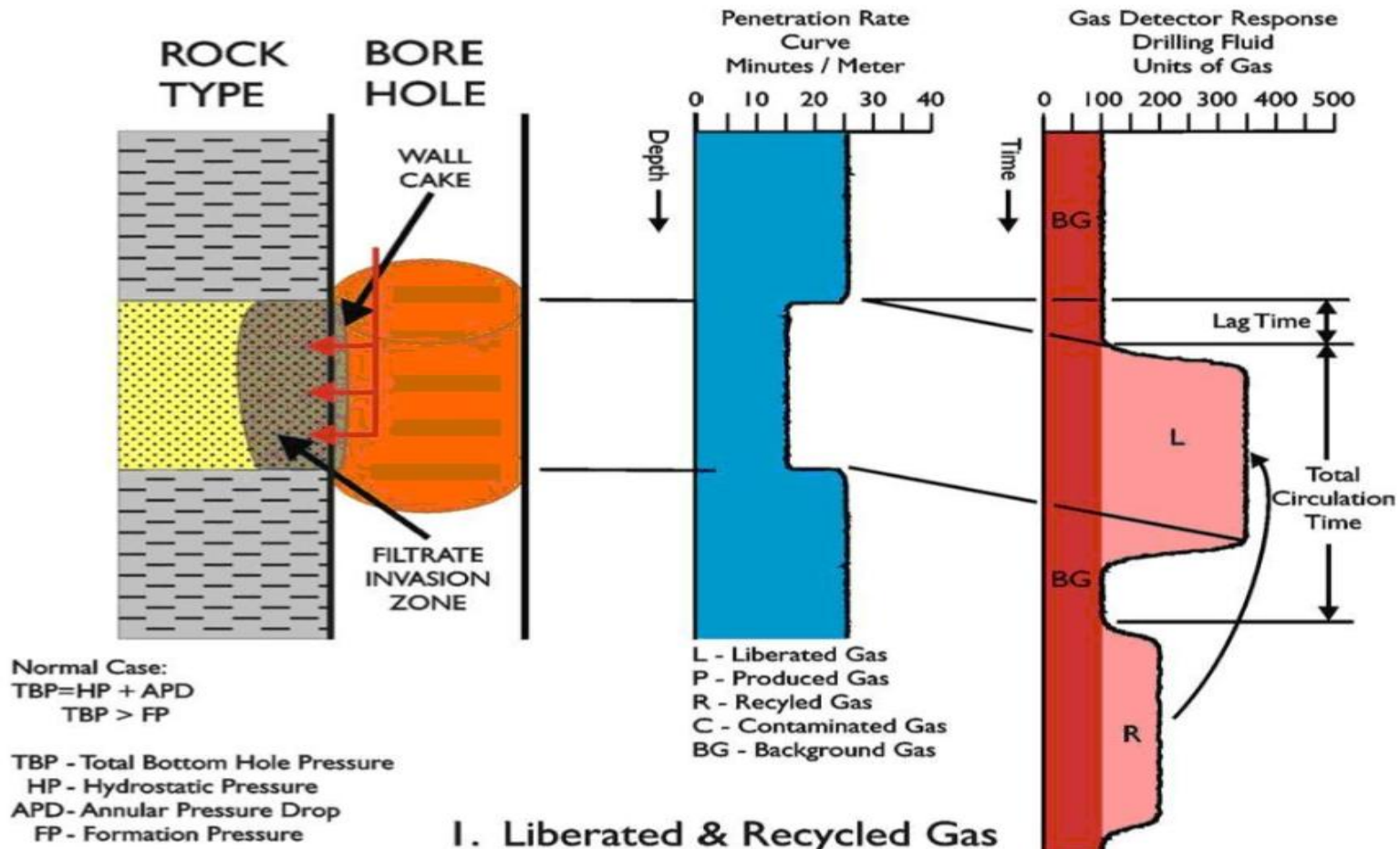


Max Gas

Background Gas

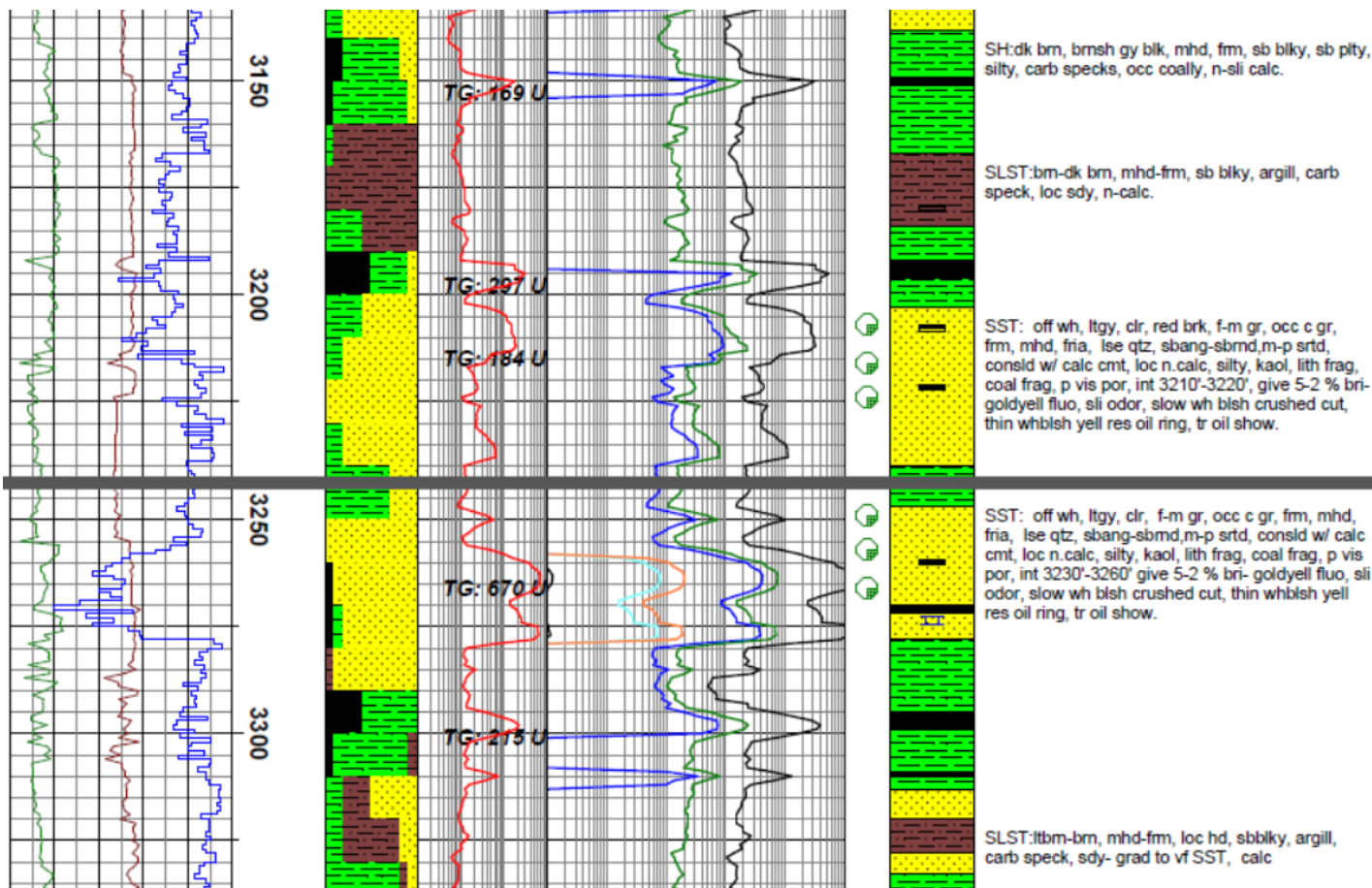
Liberated Gas

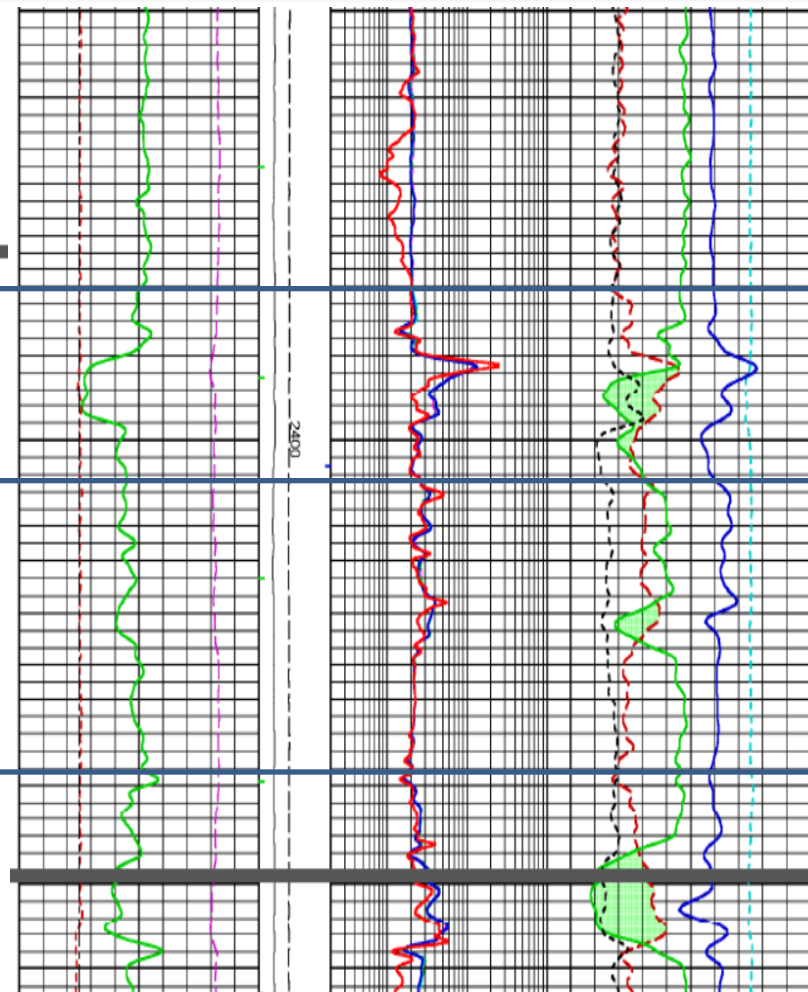
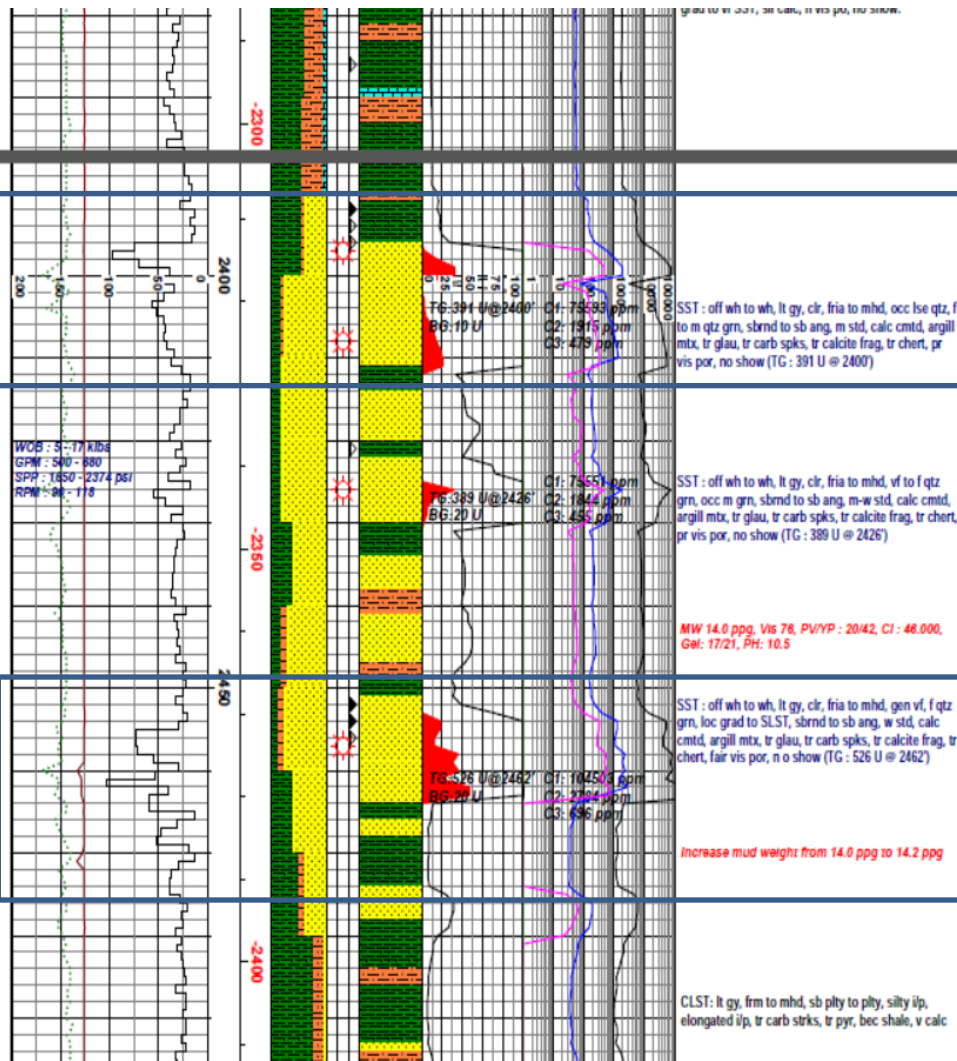
Liberated / Drilled gas : Gas didefinisikan sebagai gas yang dibebaskan secara mekanis oleh bit ke dalam cairan pengeboran ketika bit menembus formasi / gas yang terkait dengan fluida di batu pada saat pen

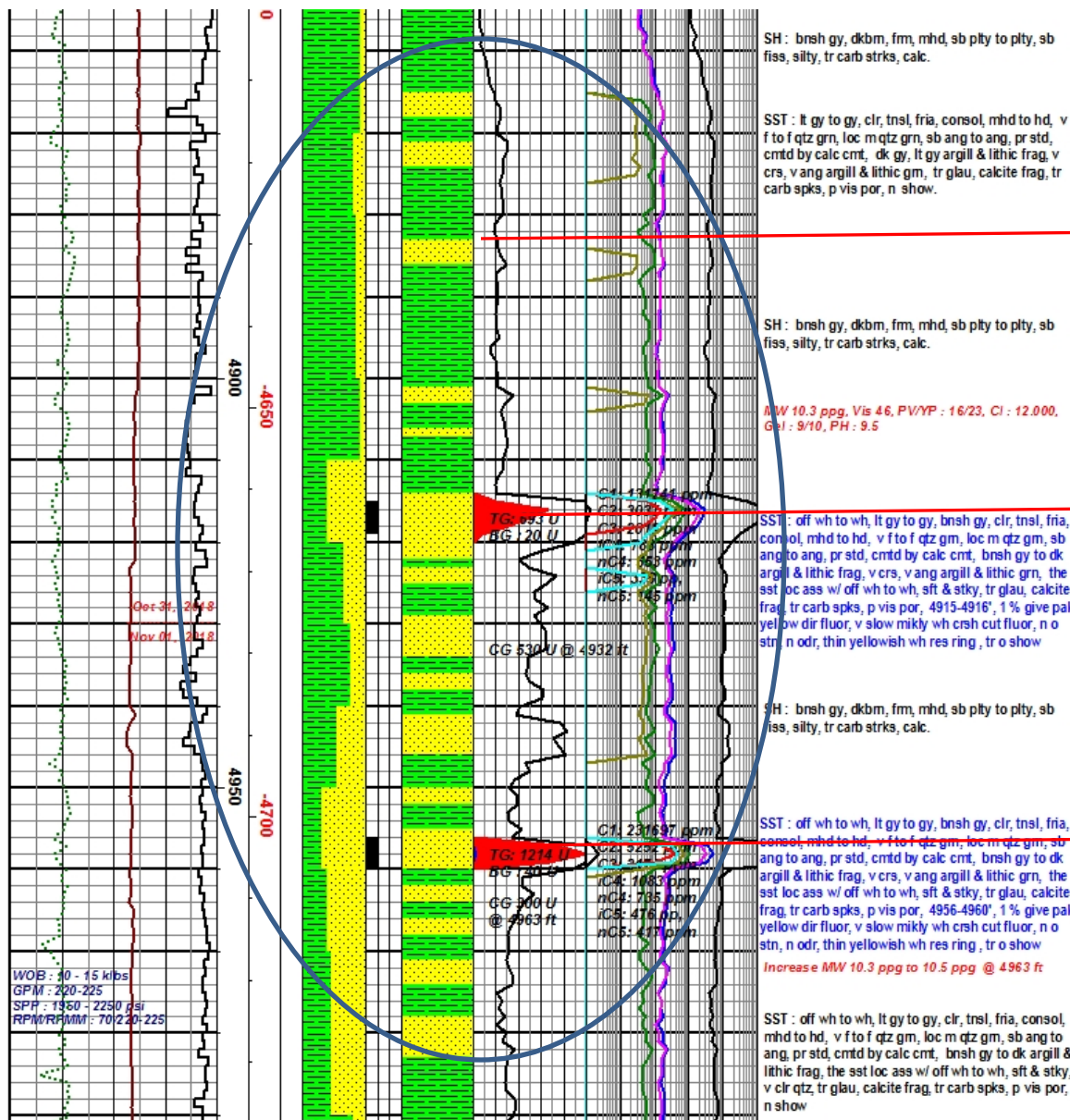


Tiga prinsip dasar penafsiran muncul dari diskusi tentang Liberated Gas

1. Pada respon gas tipe satu (mis. Di mana tekanan formasi kurang dari total tekanan lubang dasar) hanya gas yang dibebaskan secara penetrasi yang membentuk show.
2. Konfigurasi gas adalah indikasi awal dari ketebalan interval pembebasan dan mungkin **kualitas porositas** serta kedalaman dan ketebalan interval yang paling berpori.
3. Kehadiran respon gas tipe satu tidak memberikan indikasi langsung ada atau tidak adanya permeabilitas. Jika tidak ada **permeabilitas** yang efektif saat mengebor zona hidrokarbon, gas yang dibebaskan masih akan terjadi. Jika permeabilitas hadir dan kelebihan hidrostatik (PH) yang cukup dibawa dalam sistem lumpur, konfigurasi gas yang dibebaskan akan tetap relatif sama.



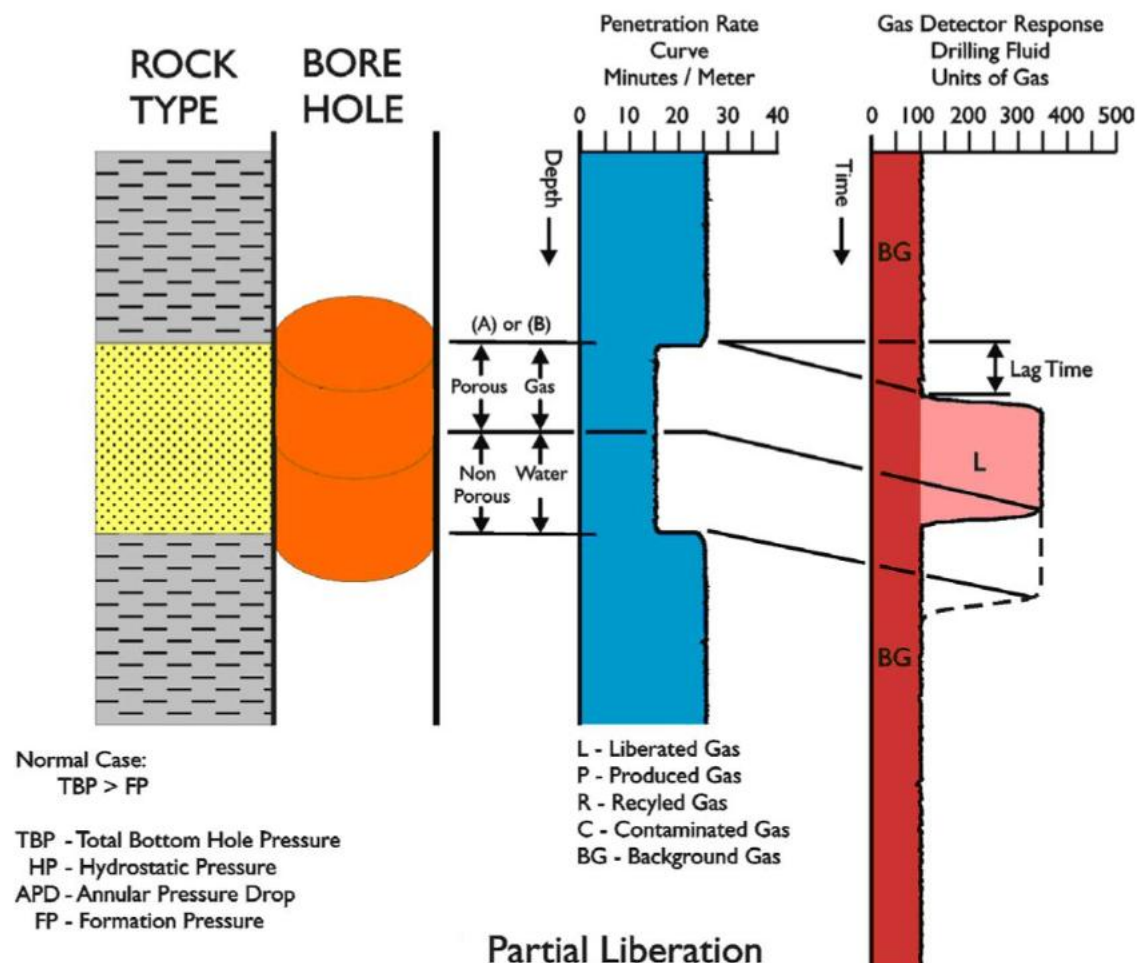


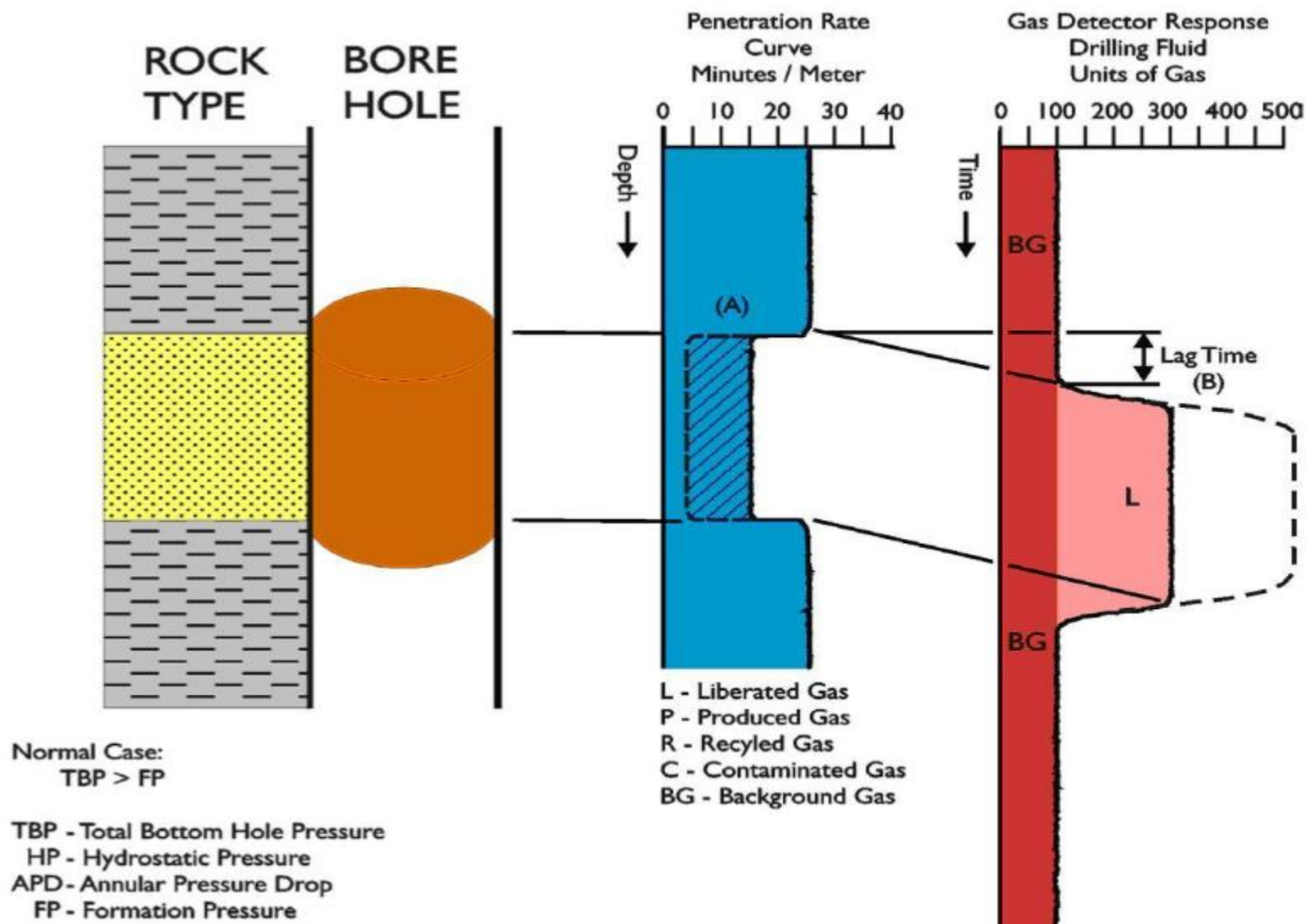


Background Gas

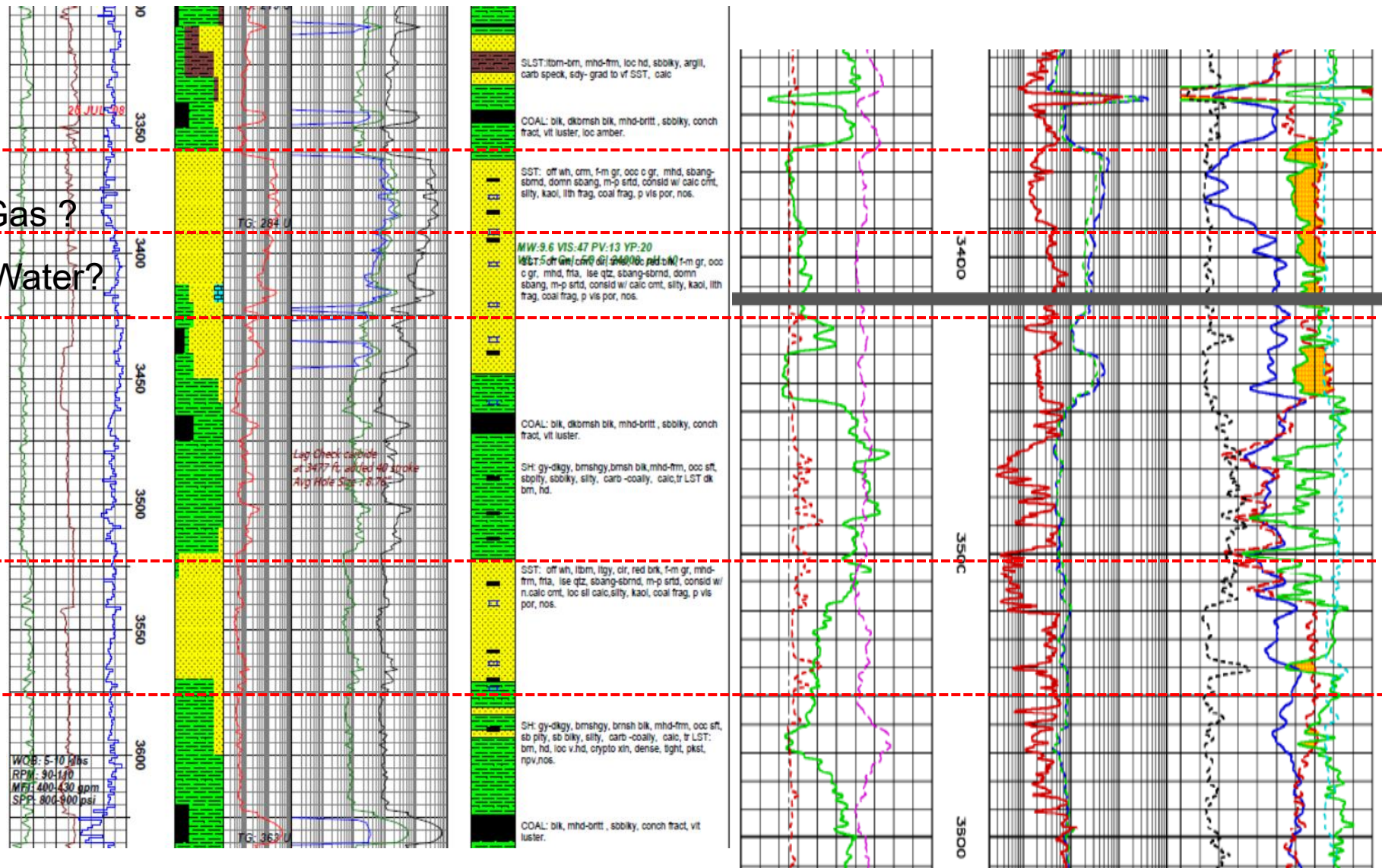
Max Gas / Librated Gas

Max Gas / Librated Gas





Gas ?
Water?



Recycled Gas

Jika gas lumpur tidak sepenuhnya menguap di lubang pengendapan tetapi dipompa kembali ke lubang, detektor gas dapat merekam penampilan kedua dari pertunjukan yang sudah ada sebelumnya. Fenomena ini digambarkan dalam gambar satu di mana pertunjukan gas yang dibebaskan telah Recycled ke permukaan untuk kedua kalinya dan ditunjuk sebagai R.

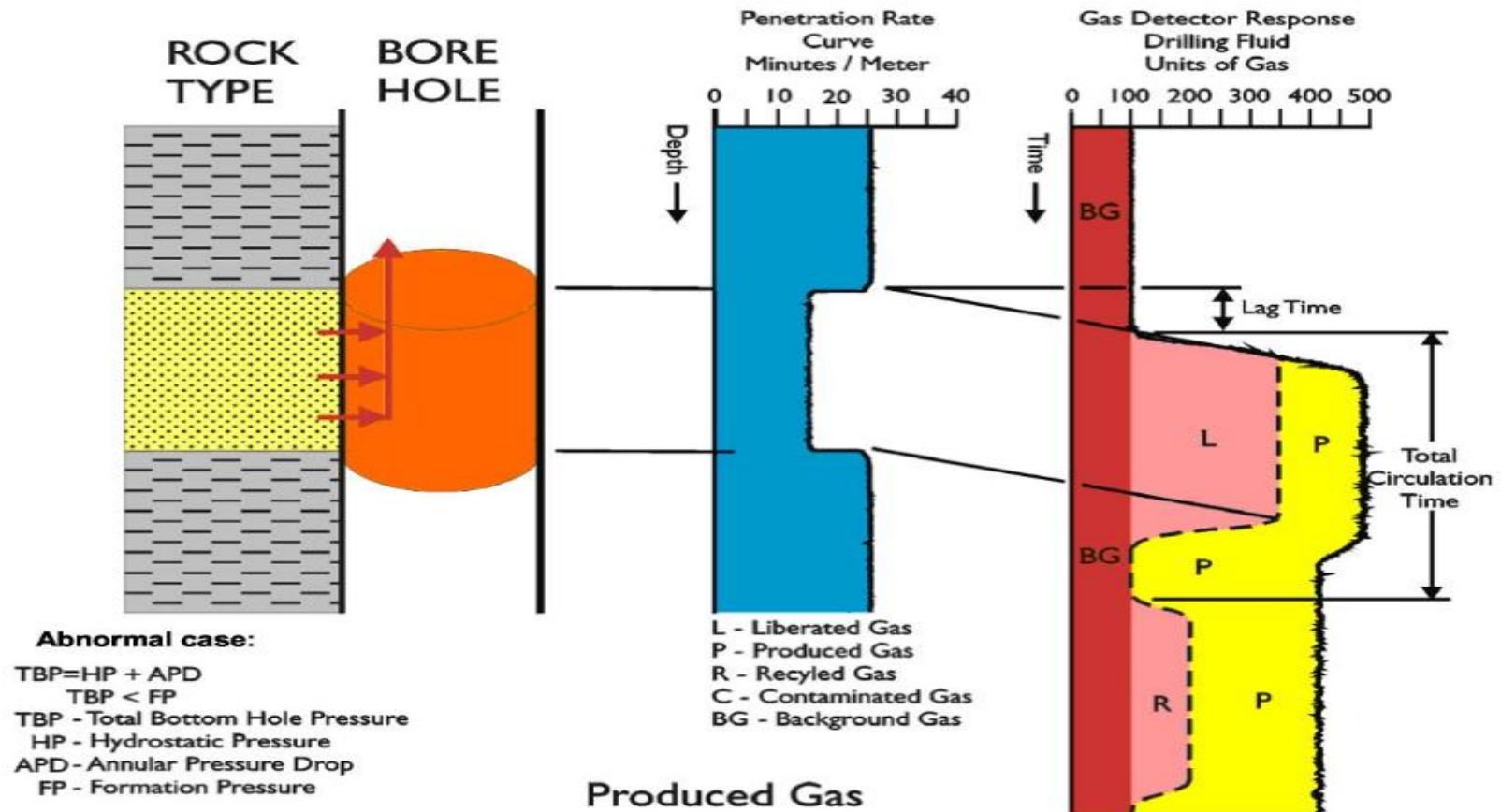
Oleh karena itu gas recycled didefinisikan sebagai gas yang telah dipompa kembali ke lubang untuk muncul kedua kalinya di permukaan.

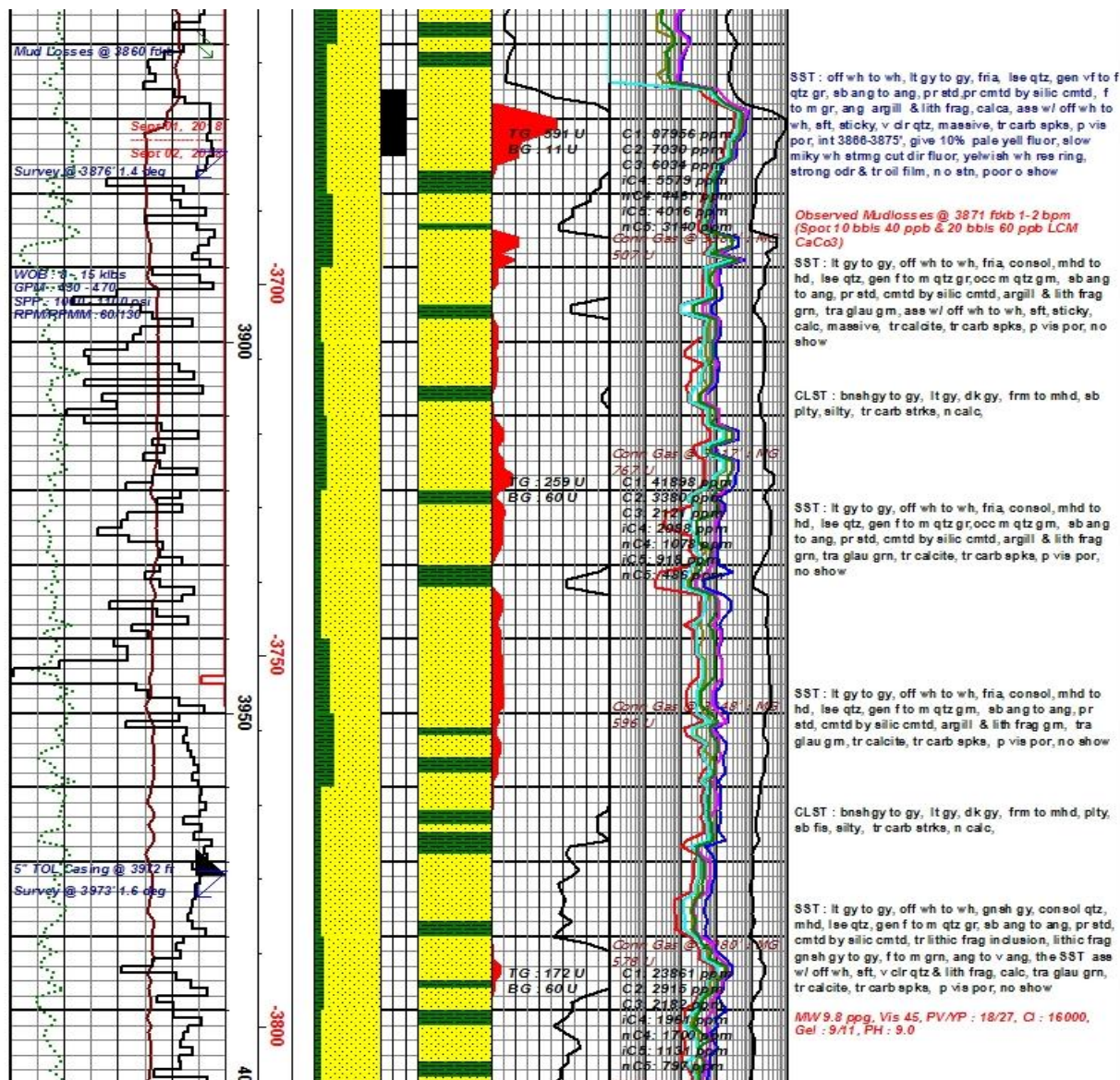
Recycled Gas dapat diidentifikasi dengan penerapan tes tertentu. Recycled harus tidak lebih besar dari respons asli tetapi harus memiliki bentuk yang serupa. Komposisi respon recycled mungkin menyesatkan karena hidrokarbon yang lebih mudah menguap sering dilepaskan ke atmosfer di dalam lubang dan di bawah pengaruh degasser. Hasilnya adalah analisis respon recycled menunjukkan proporsi yang lebih besar.

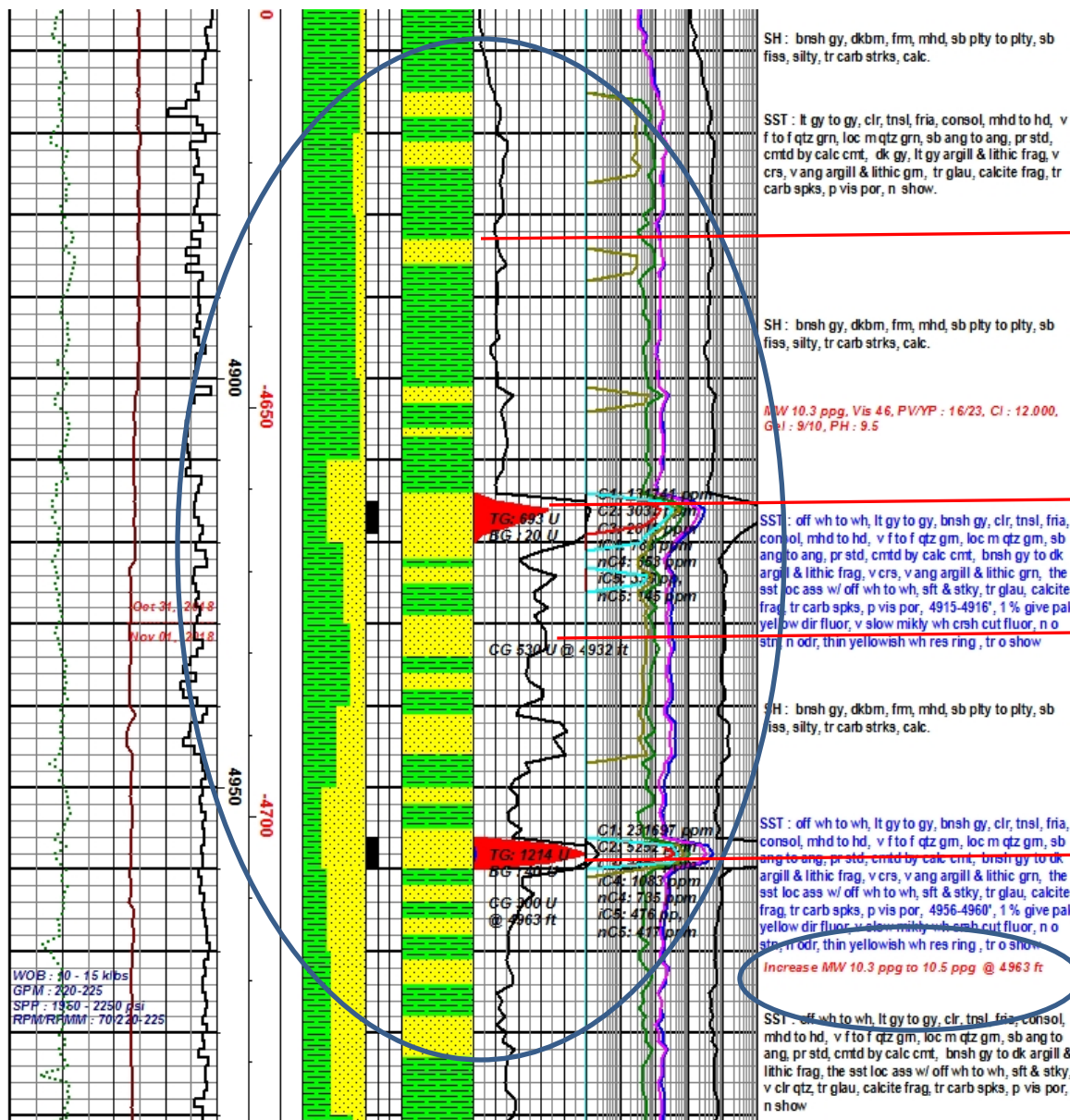
Dari awal respons gas primer hingga awal respons gas recycled dalam waktu sirkulasi adalah indikasi yang baik dari total waktu sirkulasi sistem lumpur. Informasi langsung semacam itu sering kali dapat membantu dalam memastikan keakuratan perkiraan waktu jeda.

Produced Gas

Gas yang dihasilkan karenanya didefinisikan sebagai gas yang diproduksi ke dalam fluida pemboran dari zona hydrocarbon sebagai respons terhadap tekanan formasi yang melebihi tekanan hidrostatik ($PF > PH$)





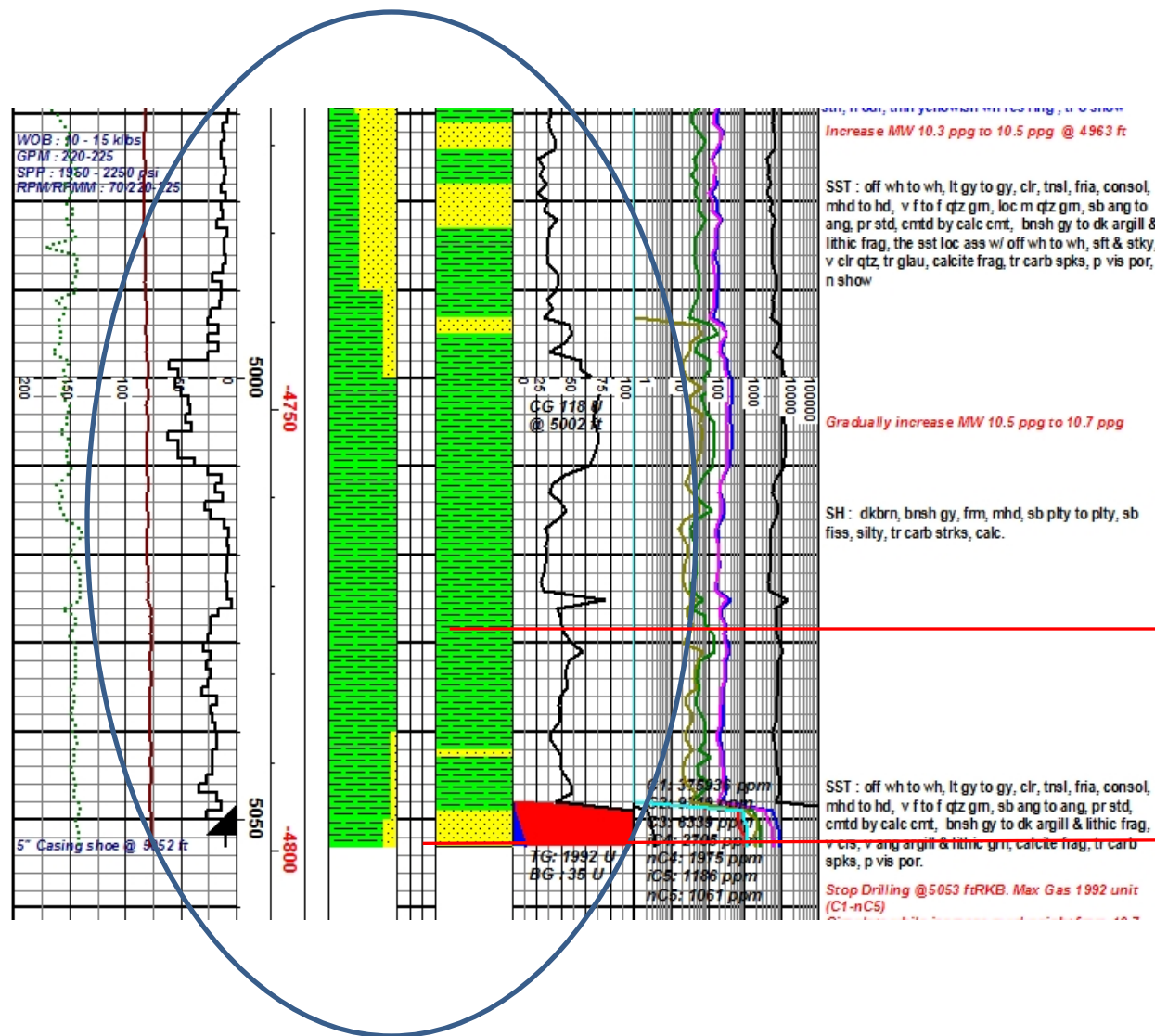


Background Gas

Drilled Gas/Librated Gas

Produced Gas

Drilled Gas/Librated Gas



Increase
Background Gas

Drilled Gas &
Produced Gas

Background Gas: Merupakan gas rata-rata yang muncul selama pengeboran menembus claystone atau shale. Bisa saja dari sandstone tetapi tidak mengandung gas hydrocaron.

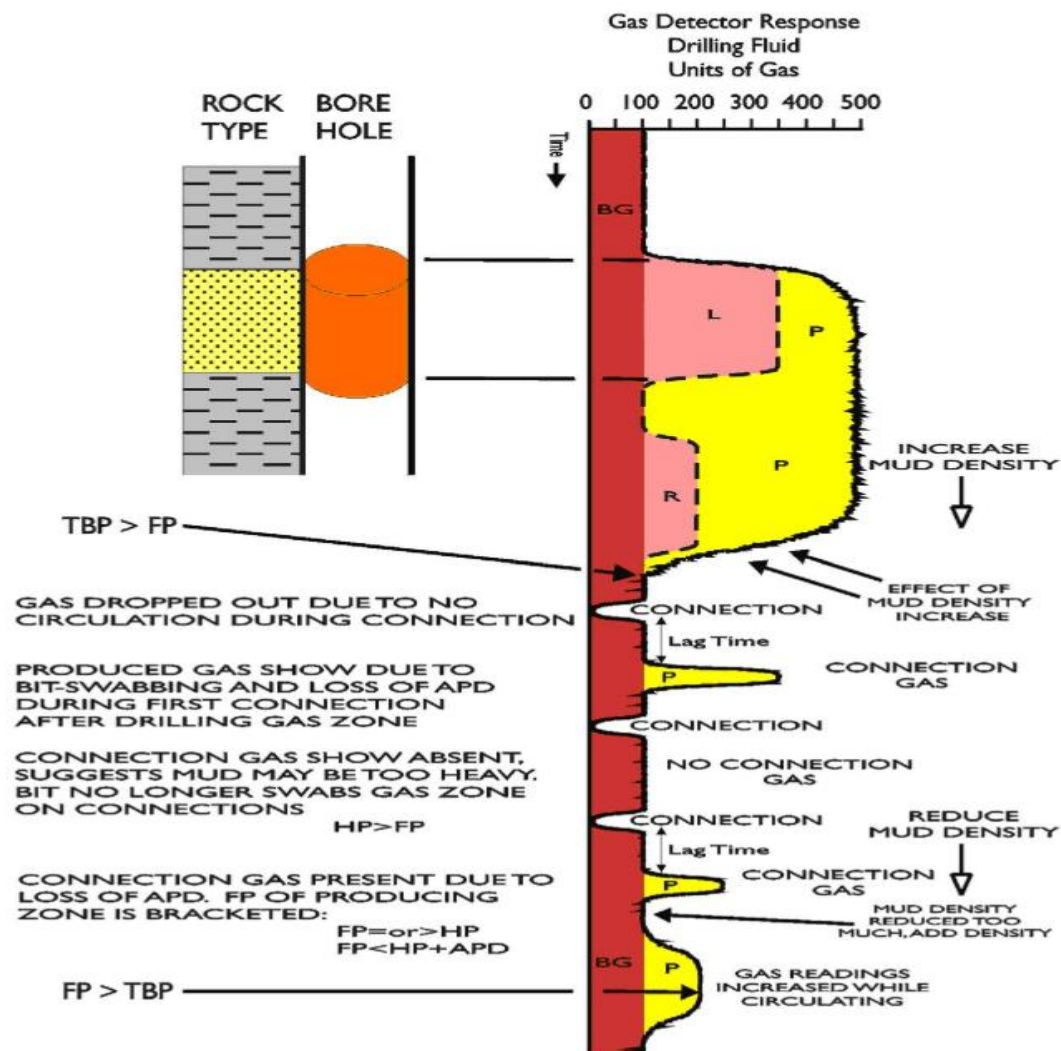
Maximum Gas : Merupakan gas terbesar diantara background gas yang muncul selama pengeboran.

Connection Gas: Gas yang muncul melebihi background gas secara significant, setelah satu kali bottom up terhitung sejak mulai pemompaan setelah connection pipe. Besarnya connection gas dihitung dari selisih dengan background gas (above background gas – ABG).

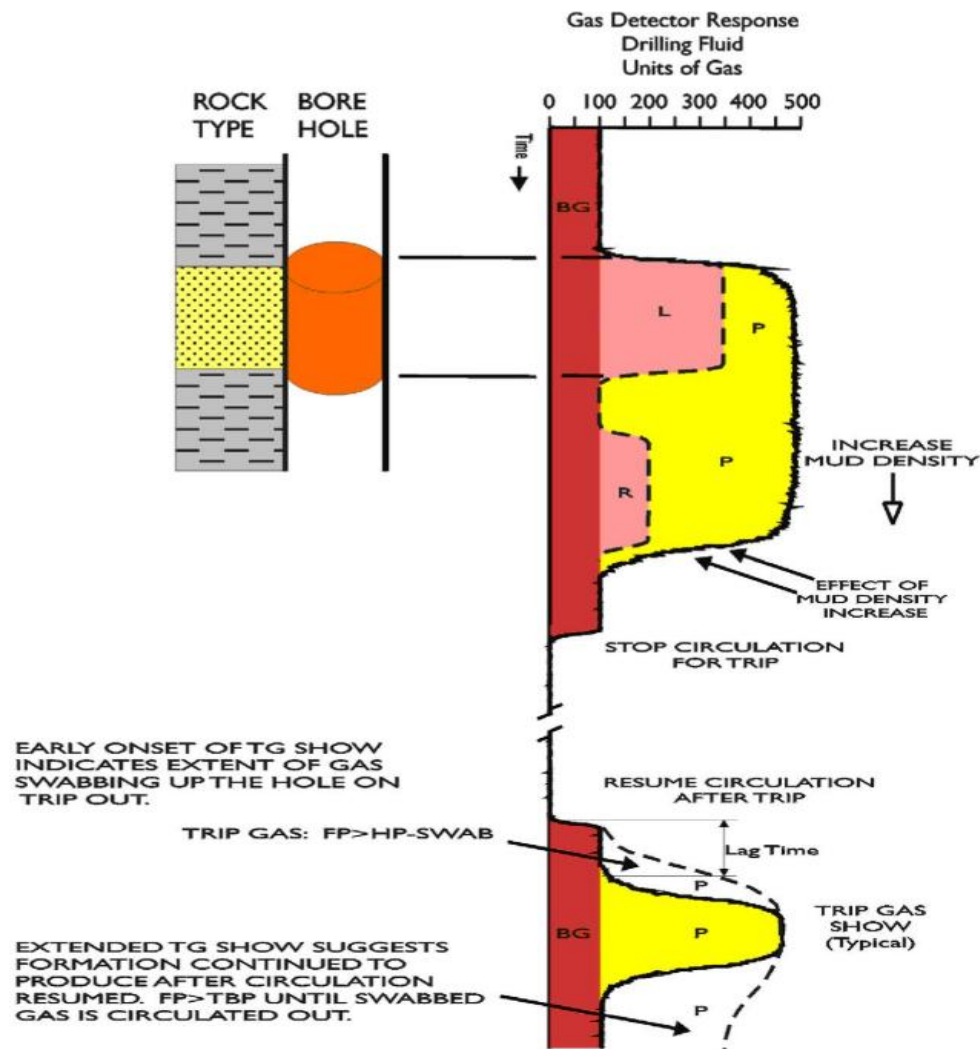
Misalnya setelah connection pompa 1 kali bottom up muncul gas 50 unit, sedangkan background gas 6 unit. Maka connection gas = 44 unit ABG. Informasikan kepada co-man bila background gas muncul 3 x connection dan cenderung naik atau tidak. Munculnya connection gas menandakan tekanan hydrostatis Lumpur sudah tidak mampu lagi menahan tekanan formasi. Tindakan preventif adalah menaikkan berat dari Lumpur.

Trip Gas : Gas yang muncul setelah satu kali bottom up terhitung sejak pemompaan saat bit mencapai dasar setelah trip in.

Swab Gas: Gas yang muncul setelah satu kali bottom up terhitung sejak pemompaan saat bit diangkat dari bottom



Abnormal Pressure Indications From Connection Gas



Blow-Out Indications From Trip Gas

DESKRIPSI	KEMUNGKINAN YG TERJADI	YG PERLU DI CHECK	AKSI/INFORMASI
Pembacaan gas 0 Unit	Lumpur yang high visc dan lumpur yg berat System gas yang tidak berjalan	Gas trap yang tidak berputar secara maksimal Sumbatan pada line, vacum pump yg tdk bekerja. System GC yang bermasalah	Reposisi gastrap Check sesuai alur dari gas trap ke GC. Check dng mix gas dari line gas trap ke GC.
Background Gas	Pembacaan minimum dari gas		
Connection Gas (Kenaikkan gas saat 1Xbottom up/Lag time setelah koneksi pipa kemudian kembali ke background gas)	Efect swab dari pergerakan pipa sebelum melakukan koneksi. Gas cut mud	Hitung theoritical dari perhitungan bottom up stroke/Lag time. MW Out Ukuran dari cutting dan d' Exsponent bila memasuki abnormal pressure	Driller/Co.man/Geologist Naikkan MW, tapi pastikan dengan paramter lainnya seperti : BG, ukuran cutting, pola D'exsponent. Catat BG & Total gas

DESKRIPSI	KEMUNGKINAN YG TERJADI	YG PERLU DI CHECK	AKSI/INFORMASI
Trip Gas (Kenaikkan gas saat 1Xbottom up/Lag time setelah trip pipa kemudian kembali ke background gas)	Efect swab dari pergerakan pipa saat melakukan trip. Gas cut mud	Hitung theoritical dari perhitungan bottom up stroke/Lag time. MW Out	Driller/Co.man/Geologist Catat BG & Total gas
Kenaikkan gas setelah 1X lag Time setelah drilling break dan kembali ke background	Menembus formasi yang poros	Cutting sample Flourescence	Driller/Geologist (sirkulasi untuk out gas sampai kembali ke BG) Catat BG & Total gas
Kenaikkan gas setelah 1X lag Time setelah drilling break namun tdk kembali ke BG tetapi makin tinggi gasnya	Memasuki zone formation gas	Pit level Mud flow out MW Out Cutting sample Flourescence Estimate PP	Driller/Geologist (sirkulasi untuk out gas sampai kembali ke BG, bila perlu naikan MW) Catat BG & Total gas
Tidak ada connection gas	MW yg tinngi atau rendahnya formasi poros		-
Kenaikkan gas relatif pelan kemudian turun (tdk ad CG & drilling break	Recycled Gas	Check dari total sirkulasi surface to surface Degasser	-

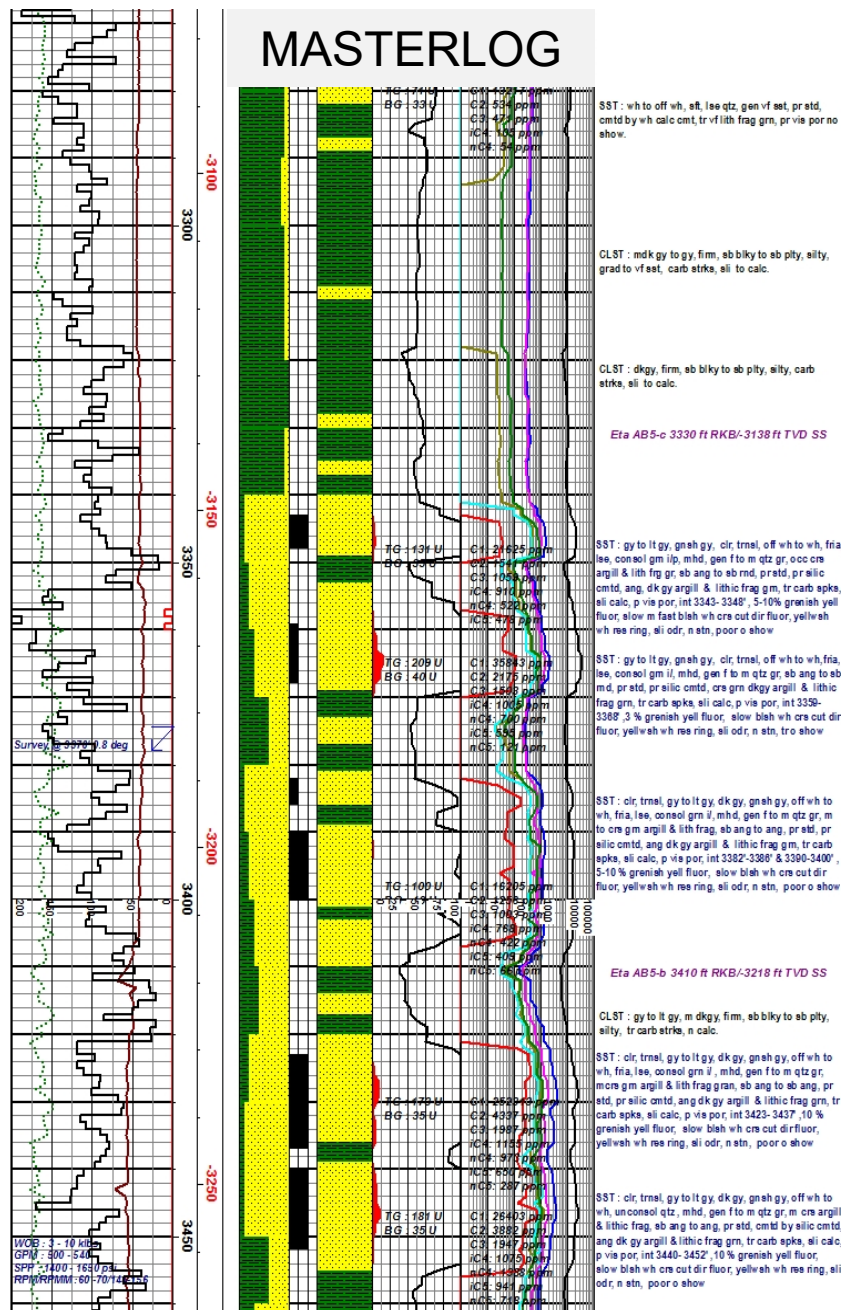
Contamination Gas

Contamination gas :This is the gas originating from the additives mixed in the active pits such as diesel, crude, different mud, chemicals, ...etc.

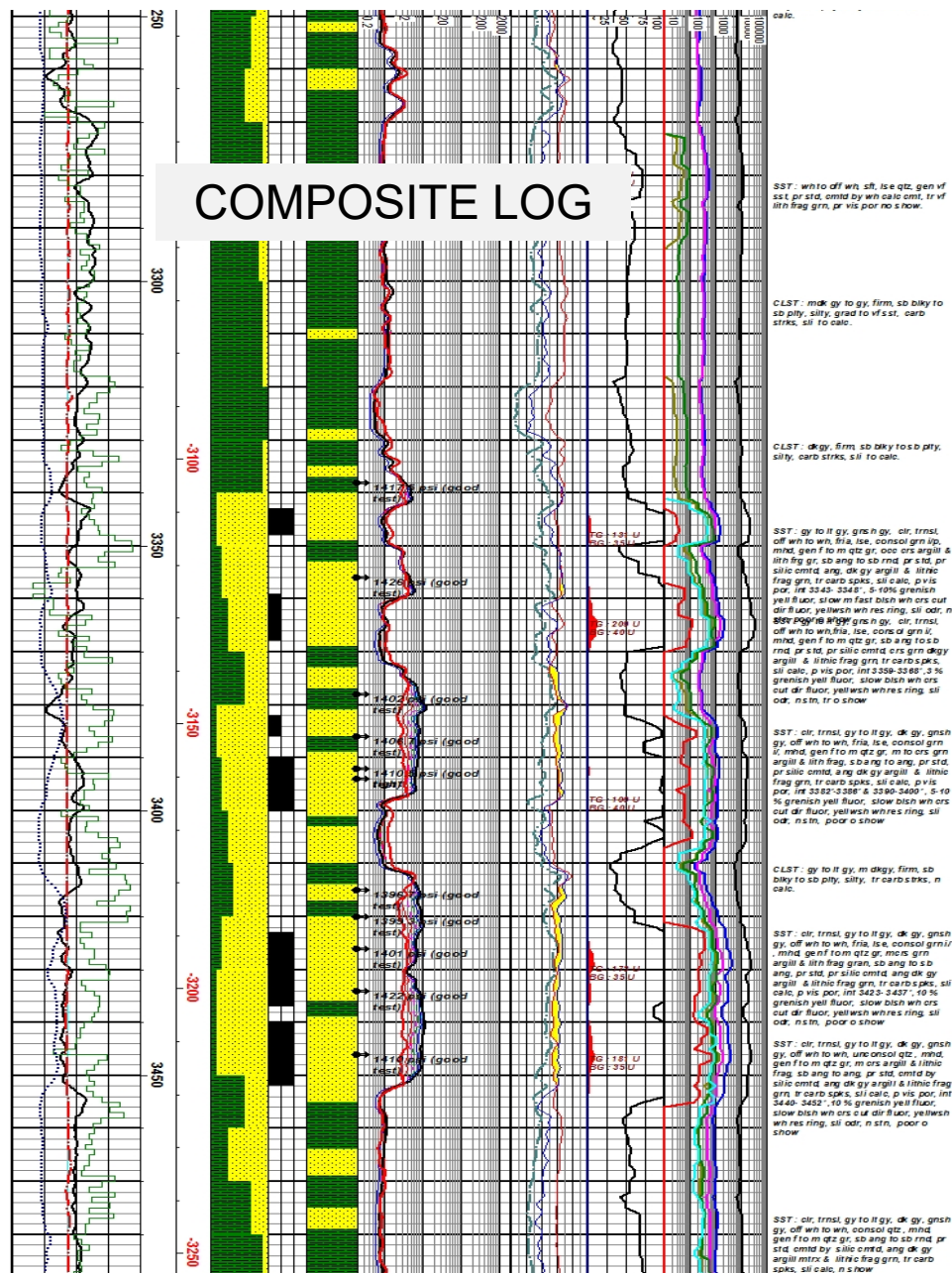
Being informed about the operations running on the pits will help identify quickly this source of gas.

Remarks : 1- Contamination gas in active will later cause recyclin
2- Oil-based muds may also be a source of contamination gas.

MASTERLOG



COMPOSITE LOG



KEYS FOR GAS ANALYSIS

Key # 1- Representativity of gas measurements

a) Degasser

- . Check that degaser tank is clean (no cutting accumulation inside gas trap)
- . Check that air intake not plugged
- . Check that suction probe grid not plugged
- . Check that mud outlet of the degasser tank is constant
- . Degaser protected from wind variations
- . Accurate position of suction probe :
 - i) It has to be as close as possible to the flowline outlet to avoid any mixing effect
 - ii) It has to be as deep as possible (without plugging it by cuttings) to recover more gas

N.B: Beware that changing the positionning will change gas absolute values

b) Gas line

- . Use of two gas lines
- . Gas lines clean of water by regular purges
- . Constant flow of gas monitored by pump panel pressure

c) Gas device using FID detectors

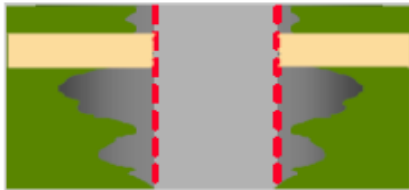
- . Check stability of recommended parameters of flow, pressure and temperature
- . Regular check of chromatograph windows
- . Repetitively of calibrations

d) Lag time validation

- . Lag time must be accurate. Check it from time to time, especially if cavings are suspected

LAG TIME PROBLEM

Lag time observed > lag time theoretical



“Cuttings arrive «late» regarding to ROP, **Hole enlargement**=>Caving, Washout, reducing hole cleaning efficiency.”

Lag time observed < lag time theoretical



Cuttings arrive «early» regarding to ROP, **Tight Hole**=> STICKY HOLE Shale hydration (Monmorillonite), ...

LAG TIME

“The correct calculation of lag time will gives a correct depth on which depth are the data coming from, then making a lag time check is so important.”

$$\text{Lag Time} = \frac{\text{Annular Volume}}{\text{Flow Rate}} \Rightarrow \text{min}$$

$$\text{Lag Stroke} = \frac{\text{Annular Volume}}{(\text{Stroke vol}) * (\text{Pump eff.})} \Rightarrow \text{stroke nb}$$

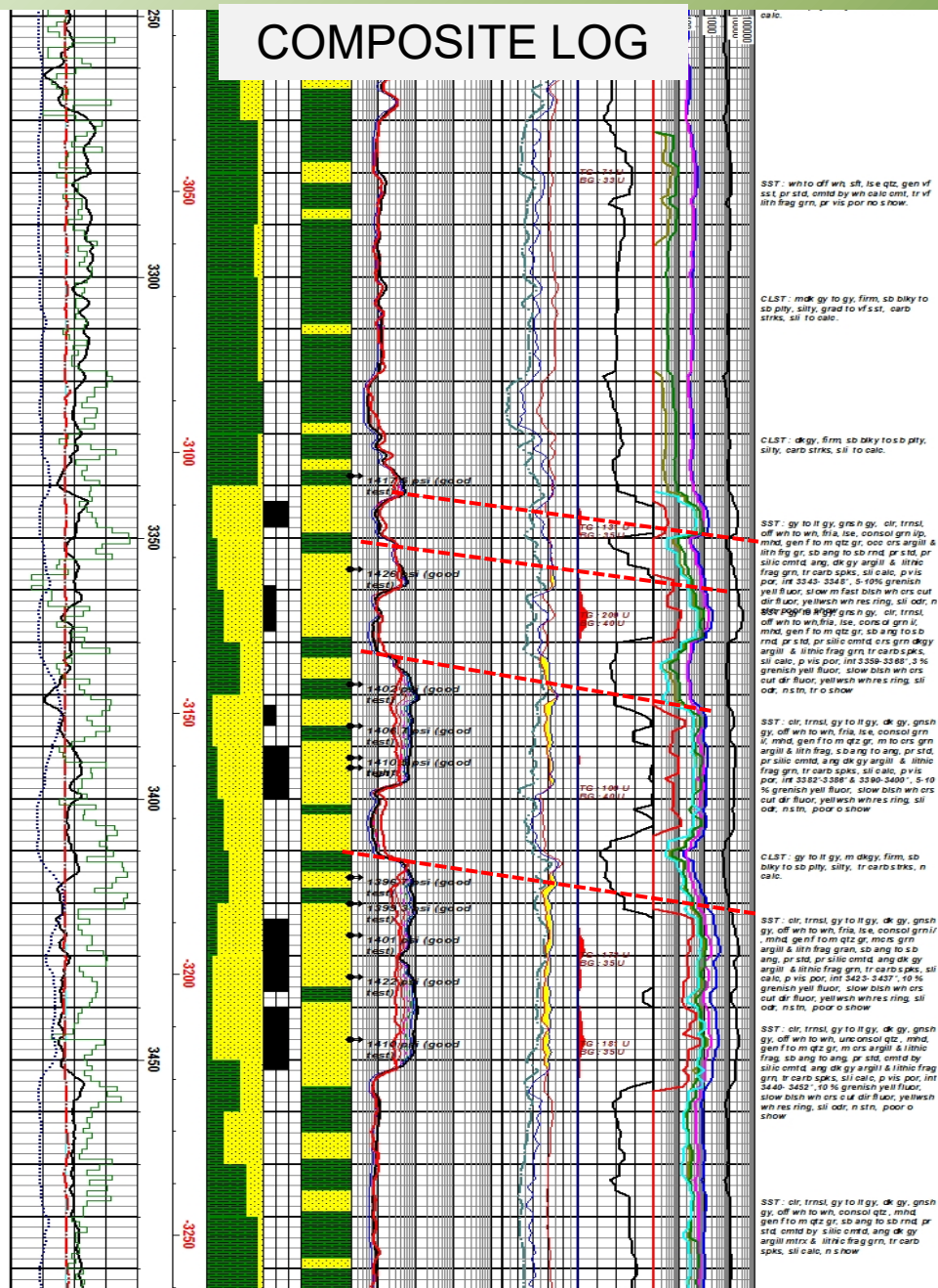
$$\text{with } AV = (\text{OH vol} + \text{ID}_{\text{CSG}} \text{ vol}) - (\text{OD}_{\text{RON}} \text{ vol in hole})$$

$$FR = (\text{Stroke vol}) * (\text{Pump efficiency \%}) * (\text{SPM})$$

BUT, IT MAY ALSO BE DUE TO:

- Wrong Pump efficiency (Toolpusher data) and/or false adjustment (Data Engineer)
- Incomplete or erroneous pipes dimensions (OD & ID) and volumes, ...

COMPOSITE LOG



Key # 2- Sensitivity of measurements - Fast-check field methods

- a) Suction of a syringe at gas line end. Observation: Total Gas spreading and dilution effect through gas line. Transit time validation.
- b) Natural degasing of mud and recycling when circulation stops.

Observation: Total Gas exponential decrease. Heavier components liberate more slowly comparatively (see scales) than lighter ones.

Key # 3- Efficiency of degasser

Using the VMS (Vacuum Mud Still) device we know the proportion of gas analysed and gas really present inside the mud. Ideally, The efficiency will vary according to several parameters such as temperature, mud rheology, ... etc

Key # 4- Continuous follow-up

Continuous follow-up will be very little time-consuming and will avoid erroneous diagnostics. Keep in mind that :

1. Any attempt to look at gas data cannot be separated from operationnal context.
2. To detect any PRODUCED or RECYCLED gas it is necessary to compare with standard conditions : a gas associated to the drilled rock volume (DRILLED GAS)
3. Any deviation from standard conditions is not instantaneously observable, so a continuous follow-up by several play-back versus time and depth is compulsory.

Key # 5- When to use Depth and / or Time ?

The best method is to monitor continuously both parameters, time and depth respectively. The judicious states of observation for each parameter are, among others :

TIME : . For time intervals including circulation stop such as pipe connection, pump-off, short trip ...etc
 . For time intervals including non drilling events such as reaming, circulating off-bottom ...etc
 . For time intervals with high ROP
 . For time intervals including a drilling break

Remark : Since the database time is overwritten, the monitoring has to be done on a short-term basis.

DEPTH : . For continuous follow-up through drilled intervals
 . For an observation in parallel to lithology
 . For a bulk overview by increasing progressively depth intervals

Key # 6- Operational context

The interdependence of these parameters is too complicated and we'll consider only the most important by categories, among which :

NATURE OF OPERATION

- a)Phase diameter change
- b)Coring
- c)Drilling mode (rotary or sliding)
- d)Bit type (tricone or PDC)
- e)Bit change (same type)
- f)Additions in active pits (mud, crude (DST), diesel...)
- g)Rig degasser on / off
- h)Horizontal drilling
- i)Tripping and Reaming

How it affects gas values

Volume of rock drilled
 Volume of rock drilled
 Different kind and size of cuttings
 Different kind and size of cuttings
 Drilling efficiency (size of cuttings)
 Contamination gas - Recycling
 Degasing in pits - Recycling
 Recycling in drains
 Drilled and produced gas

LITHOLOGICAL PARAMETERS

- a)Matrix porosity
- b)Saturation
- c)Fractures / Permeability
- d)Formation pressure
- e)Oil-water transition
- f)ROP / Dexponent trend

How it affects gas values

Drilled gas increase with porosity
 Drilled gas increase with saturation
 Local production of gas
 Local production of gas
 Less gas and increase percent of recycling
 Lithological or formation pressure change

DRILLING PARAMETERS

- a)ROP
- b)FLWpmps
- c)WOB, SPP
- d)TRQ, RPM

How it affects gas values

Volume of rock drilled function of ROP
 Dilution and ECD (delta P annular) effect
 Change travel time inside pits - recycling
 Flushing into formation
 Drilling efficiency, bit wear

MUD PARAMETERS

- a)Mud type (oil or water)
- b)Density change
- c)Viscosity change
- d)Temperature at surface

How it affects gas values

More recycling with water-based muds
 Flushing, fracturation and/or production
 Degazing drops with viscosity increase
 Flushing into formation (SPP increase)
 Degazing increases with temperature

Key # 7- Pertinent states and events of observation

SENSIBILITY TO ROP

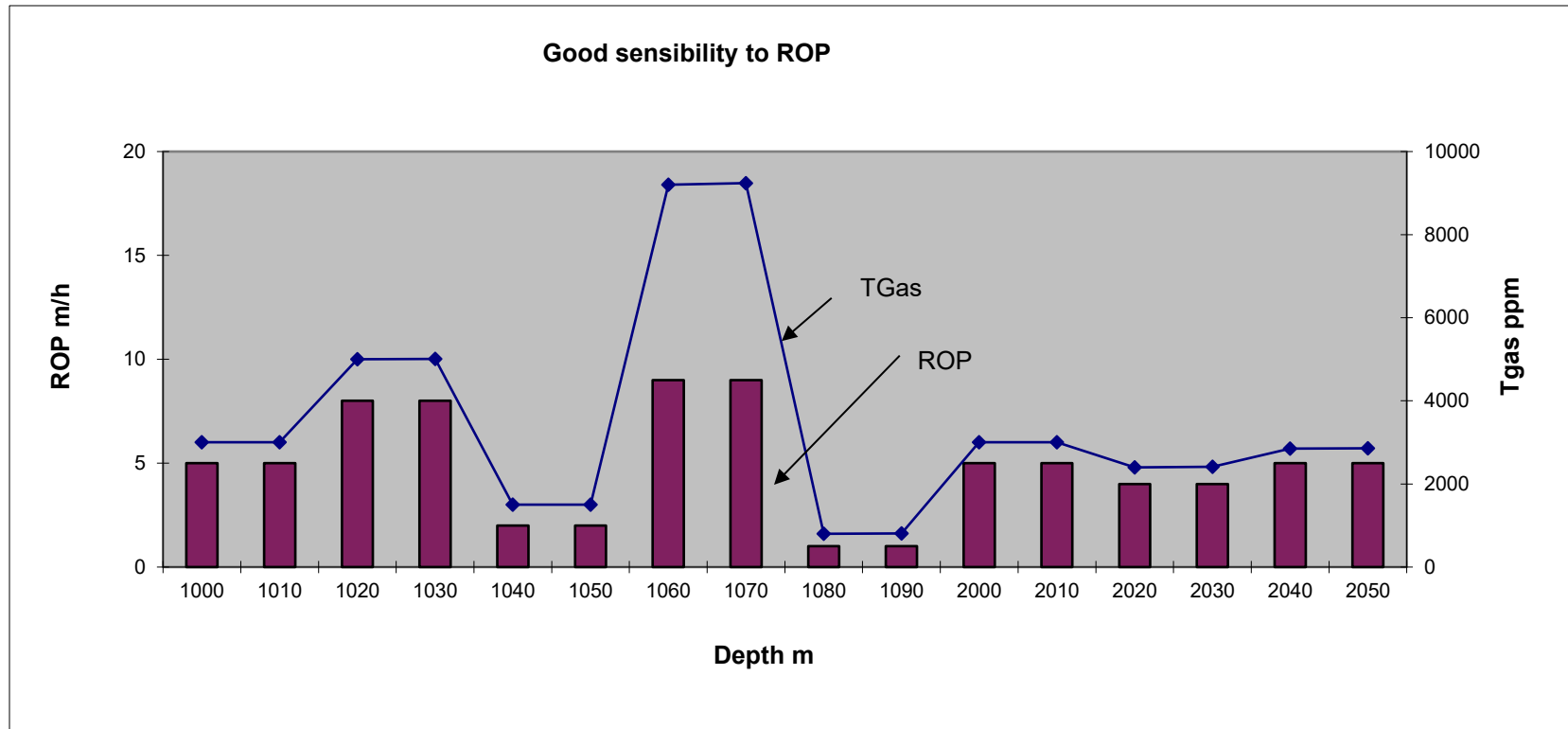
Peaks of ROP correspond to gas peaks

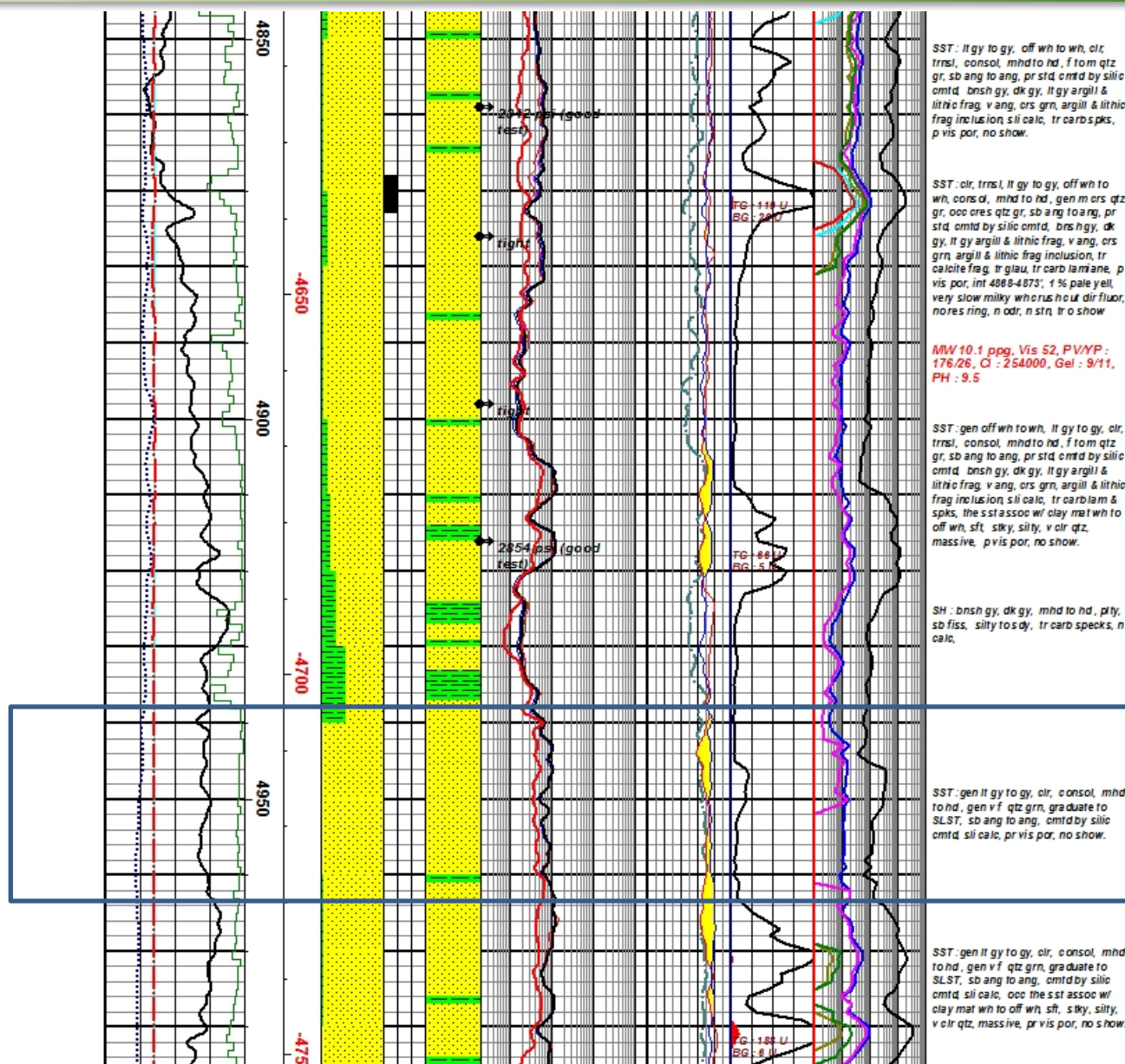
Characterization of

Drilled gas

Where to look at ?

ROP & Gas versus depth





SENSITIVITY TO FLOW

Characterization of

Where to look at ?

On bottom

Change of flow
correspond to
change of gas value ?

Yes

Drilled + Produced gas

FLWpmps & Gas (+Lagtime) vs Time

N

Recycled gas

Off bottom

Change of flow
correspond to
change of gas value ?

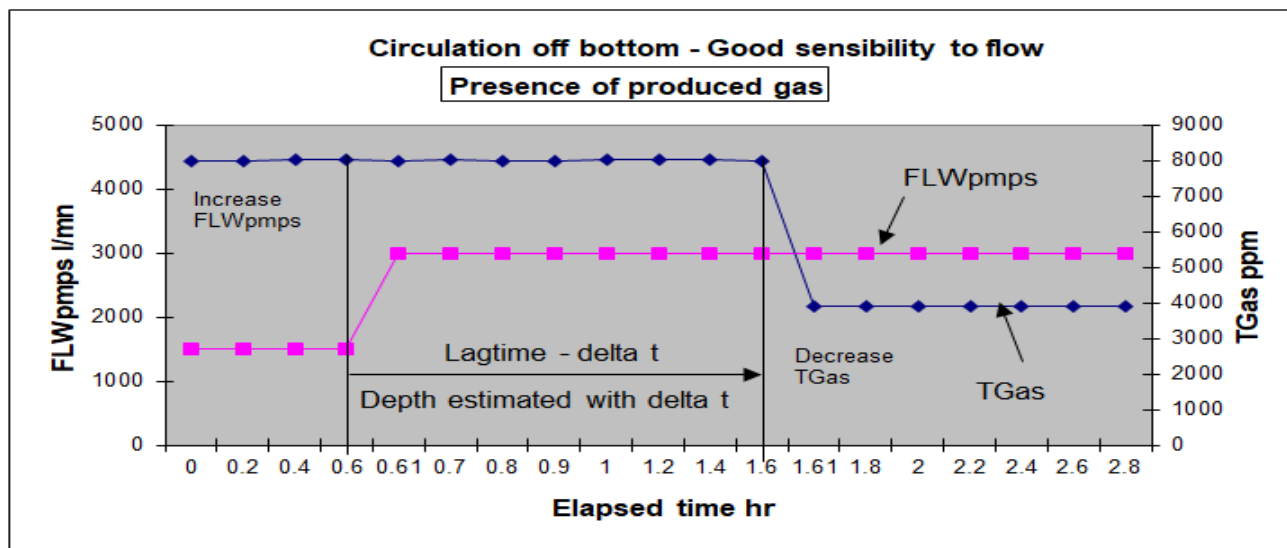
Yes

Produced gas

FLWpmps & Gas (+Lagtime) vs Time

No

Recycled gas



PIPE CONNECTION / PUMP OFF / SWAB GAS

When stopping circulation during connection or any other event, pressure losses are suppressed and when pulling pipes a swabbing effect is created. The induced reduction of pressure can favor local flow from formation. So, they can be indicative of PRODUCED GAS.

TRIP GAS

During trips, several processes can occur :

- a) Cuttings accumulation at bottom of the hole leading to DRILLED GAS
- b) Reduction of pressure (same explanation as in 5.3) leading to PRODUCED GAS
- c) Temperature decrease at surface if circulation has been stopped for a long time. This can cause a decrease in degassing efficiency (more important for heavier components) and leading to the presence of RECYCLED GAS.

Hydrocarbon Gas Data Analysis

Hydrocarbon Indication

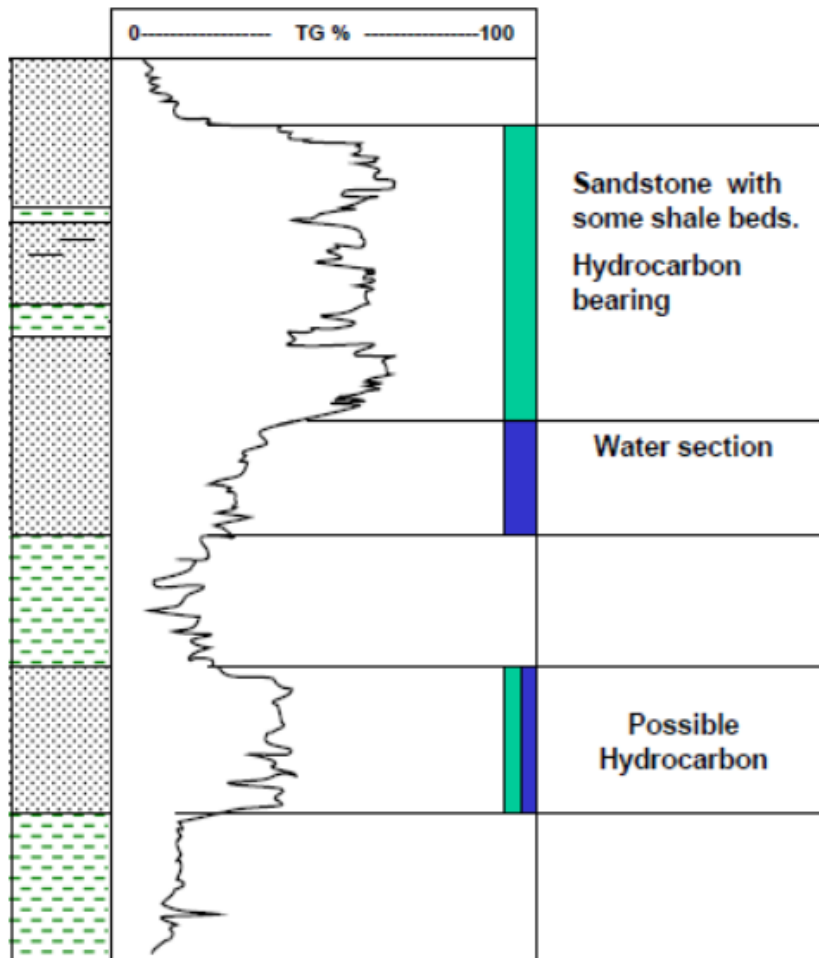
- Gas data can be used to get two basic indications of the well under drilling.
 - 1 . Saturation
 2. Type of fluid oil or gas.
 3. Other Indications are not discussed here.

Saturation : Percentage of oil or gas in the formation. Usually there is water in the formation. Oil and gas occupies the pore space besides water. Knowledge of percentage of hydrocarbon is essential. Formation with 20% Oil & 80% water will produce water only and a little oil if at all.

Hydrocarbon Indication

- Total Gas or calculated total Gas gives good idea if the bed contains any commercial hydrocarbon.
- $CG = (C1 + C2 + C3 + (iC4 + nC4) + (iC5 + nC5))$
- Normalized total gas or normalized CG can be used.

GAS DATA INTERPRETATION



Green = oil
Blue = Water

Fluid Identification There are various methods available.

1. Single Depth Pixler Plots, Triangular plots
2. Continuous Depth plots : Wh, Bh, Ch
3. Continuous Depth plots : LM, LH, HM

1. Pixler (Baroid) Method

This is perhaps the oldest of the methods described here, based on work by Pixler (1969) and Ferrie (1981). A 'Pixler' chart is part of the gas composition diagram.

The ratios used are: $C1/C2$ ($C2$ = ethane)

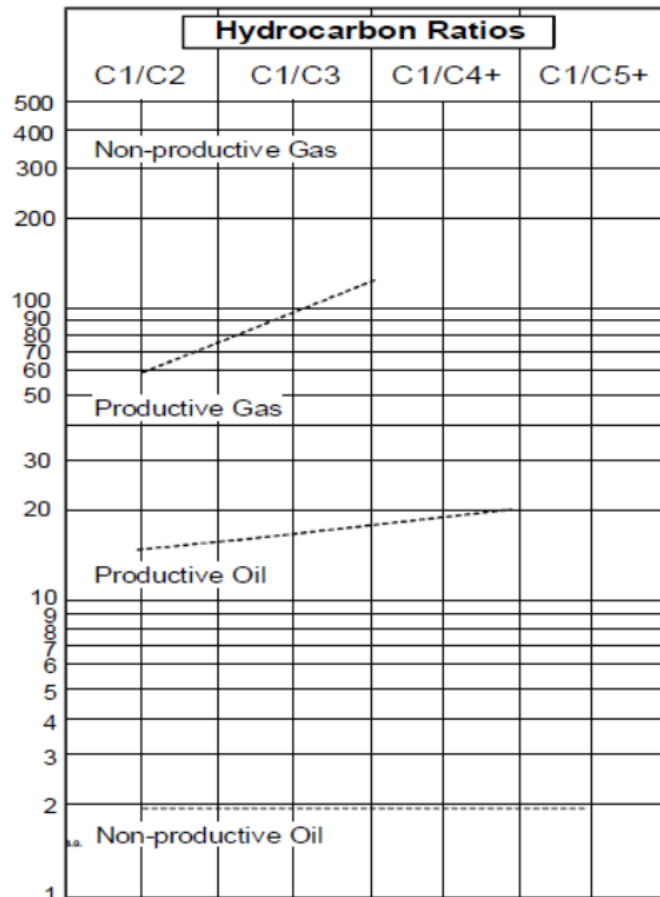
$C1/C3$ ($C3$ = propane)

$C1/C4 + C5$ ($iC4$ = isobutane; $nC4$ = normal butane)

If data exists for iso- and normal pentanes (not always gaseous at surface temperature and pressure) their ratios versus $C1$ may be plotted as well. Lines are then drawn to connect the individual ratios.

GAS DATA INTERPRETATION

Modified from Ferrie, 1981



1. A ratio of C1/C2 between 2 and 15 indicates an oil-bearing zone
2. A ratio of C1/C2 between 15 and 65 indicates gas
3. The lower the C1/C2 ratio, the richer the gas (or the lower the oil gravity)
4. A C1/C2 ratio less than 2 or greater than 65 is probably non-productive
5. The slope of the line connecting the plotted ratios is a qualitative permeability indicator. Steep slopes indicate 'tight' zones.

GAS DATA INTERPRETATION

Client GEO OIL Well BM #3
 Location SEINE SAINT-DENIS
 Formation _____
 Depth from _____ to _____
 Gas Data corrected ☒ uncorrected ☐

Gas (% or ppm)	Background	Net Increase		
		Plot 1	Plot 2	Plot 3
➡ Depths ➡		3 0 0 0 m	3 0 5 0 m	3 10 0 m
Methane (C1)	-----	2.40 %	3.13 %	1.7 %
Ethane (C2)	-----	0.11	0.32	0.42
Propane (C3)	-----	0.09	0.34	0.4
Butanes (iC4 + nC4)	-----	0.08	0.26	0.23
Pentanes (iC5 + nC5)	-----	-----	-----	0.06
Total gas %				

Plot 1

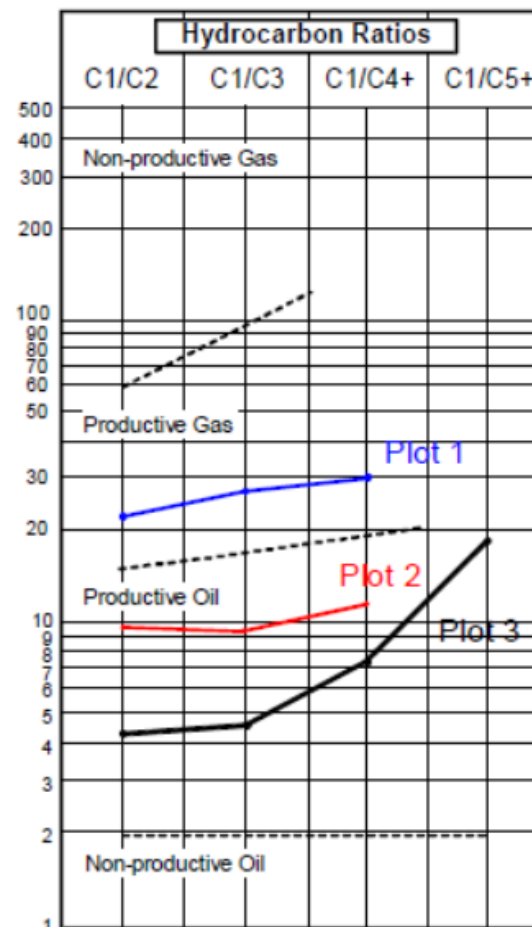
C1/C2 = 2.18 C1/C3 = 2.67 C1/C4+ = 3.0 C1/C5+ = -----

Plot 2

C1/C2 = 9.8 C1/C3 = 9.2 C1/C4+ = 12.3 C1/C5+ = -----

Plot 3

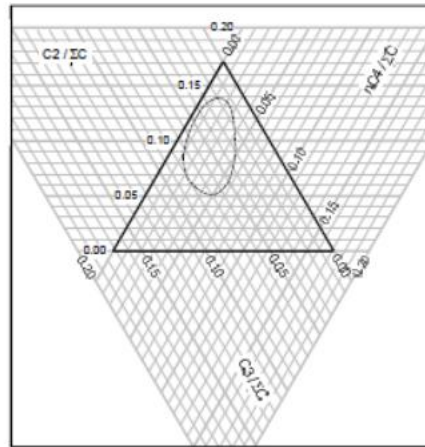
C1/C2 = 4.1 C1/C3 = 4.8 C1/C4+ = 7.4 C1/C5+ = 28.3



This method was developed within Geoservices in the early 1970's . The Triangles method plots the ratios of C₂, C₃ and nC₄ to the sum of the hydrocarbon gasses (C) for a specific depth. The plot is similar in configuration to a ternary diagram, and consists of a triangle marked within a grid. Each side of this reference triangle is defined by the minimum and maximum ratios (0 to 0.17) likely in a given reservoir. The actual ratios for a given depth of interest are superimposed on the reference triangle, as lines opposed by 120 degrees to each other. This forms a second triangle (Figure 4-7A,B), with size and orientation determined by the ratio values:

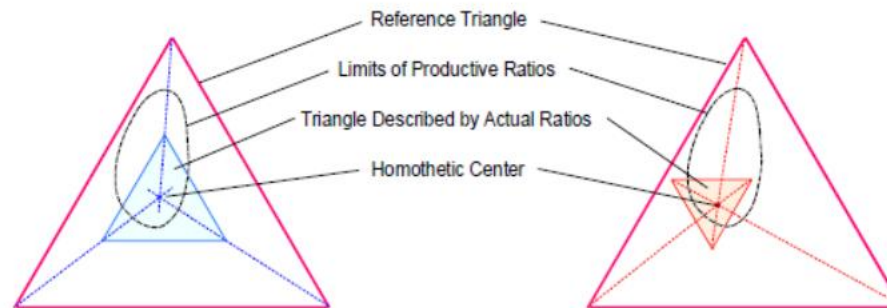
- 1. A triangle with the apex pointing upward indicates a gas-rich zone**
- 2. A triangle with the apex pointing downward indicates an oil-rich zone**
- 3. A large triangle indicates dry gas or oil with a high-Gas -oil ratio (GOR)**
- 4. A small (or nearly invisible) triangle indicates wet gas or oil with a low GOR.**

GAS DATA INTERPRETATION



A. Indicated Gas Zone

B. Indicated Oil Zone



GAS DATA INTERPRETATION

Client GEO OIL Well BM #3

Location SEINE SAINT-DENIS

Formation _____

Depth 3000m

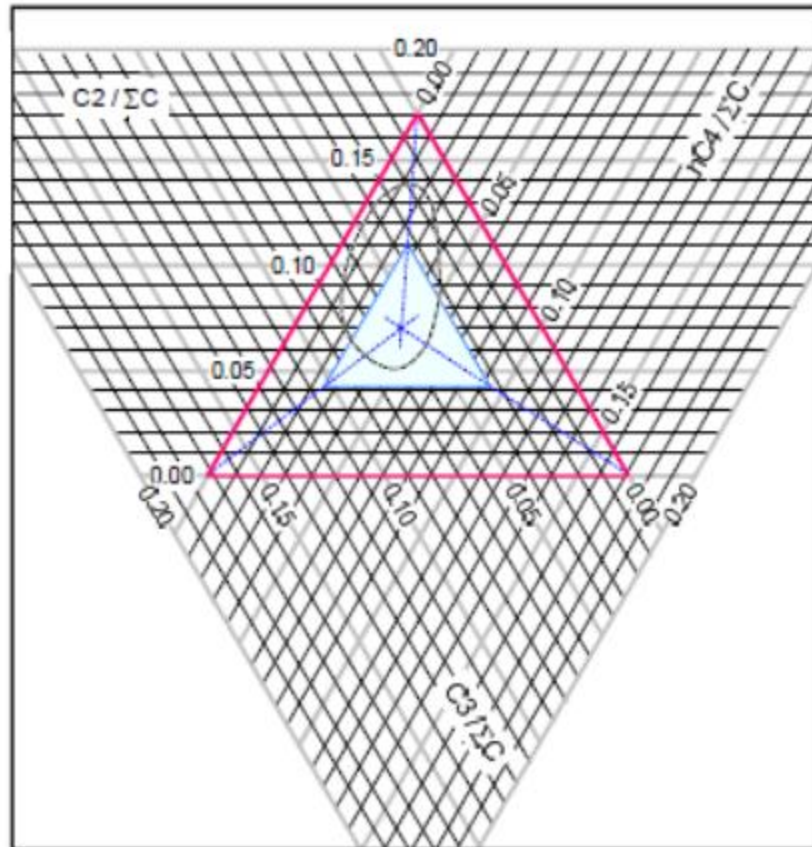
Gas Data corrected ☒ uncorrected ☐

Gas (% or ppm)	% or ppm
Methane (C1)	2.40 %
Ethane (C2)	0.11
Propane (C3)	0.09
Iso-butane (iC4)	0.02
Normal Butane (nC4)	0.06
Iso-Pentane (iC5)	
Normal Pentane (nC5)	
ΣC	2.66
Total gas %	

$$C2 / \Sigma C = 0.041$$

$$C3 / \Sigma C = 0.034$$

$$nC4 / \Sigma C = 0.023$$



GAS DATA INTERPRETATION

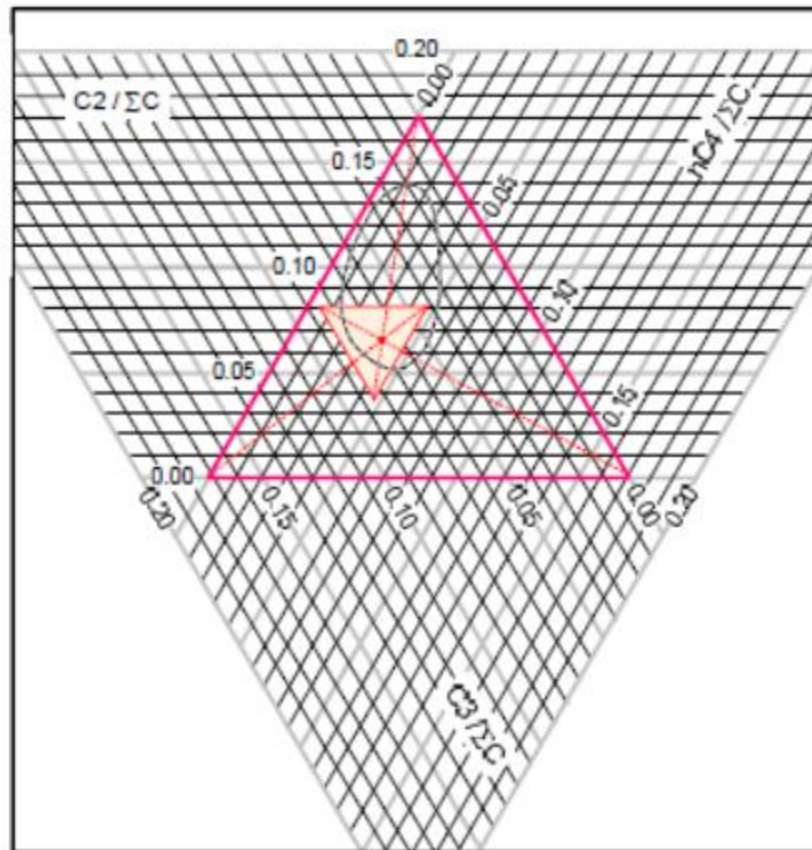
Client GEO OIL Well BM #3
 Location SEINE SAINT-DENIS
 Formation _____
 Depth 3050 m
 Gas Data corrected ☒ uncorrected ☐

Gas (% or ppm)	% or ppm
Methane (C1)	3.13 %
Ethane (C2)	0.32
Propane (C3)	0.34
Iso-butane (iC4)	0.07
Normal Butane (nC4)	0.19
Iso-Pentane (iC5)	
Normal Pentane (nC5)	
ΣC	3.98
Total gas %	

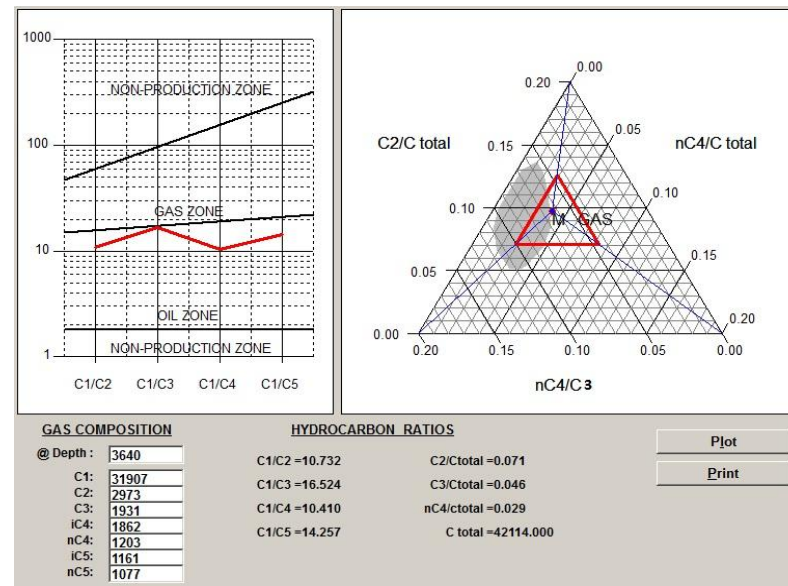
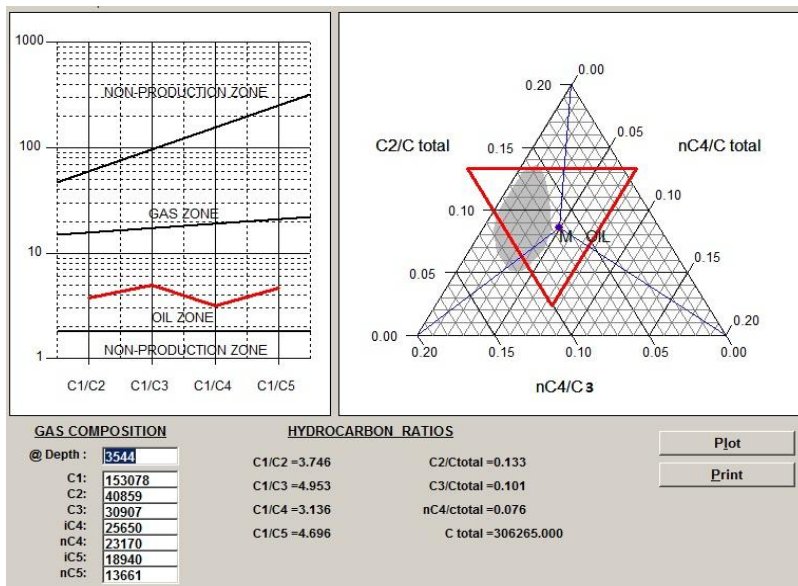
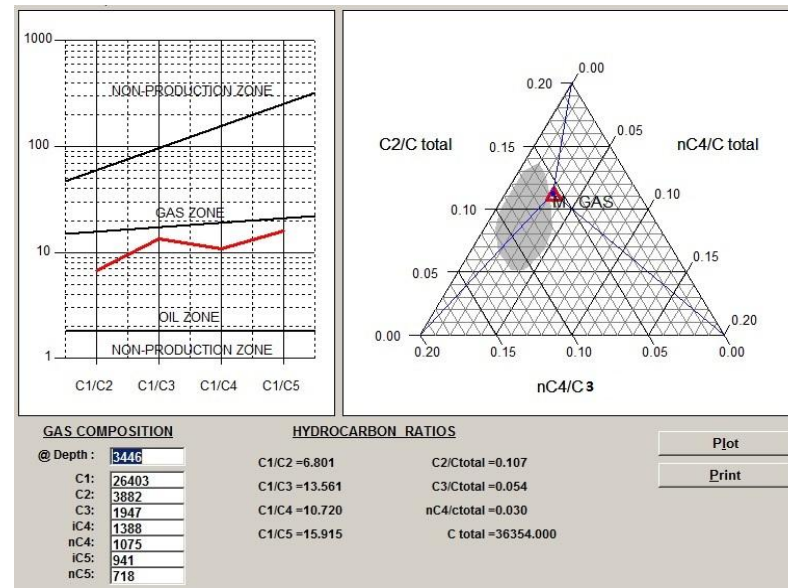
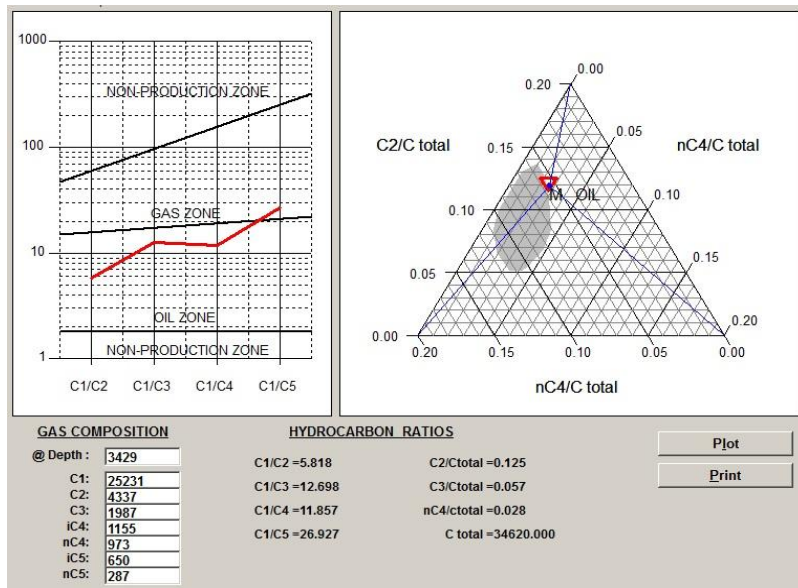
$$C2 / \Sigma C = 0.081$$

$$C3 / \Sigma C = 0.086$$

$$nC4 / \Sigma C = 0.048$$



GAS DATA INTERPRETATION



2. Continuous Depth plots : Wh, Bh, Ch

The balance ratio was selected for two reasons :

1. Bh appears on plots as nearly the nearly inverse of Wh
2. Coal-bed gas, which tends to be rich in C1 + C2, will plot differently from gas associated with oil.

According to Haworth, et al, when Wh and Bh are plotted together, fluid character may be interpreted as follows:

Bh > 100 = very dry gas

Wh in gas range and Bh greater than Wh = gas

Wh in gas range and Bh slightly greater than Wh = wet or retrograde gas

Wh in oil range and Bh less than Wh = oil

Wh greater than 40 and Bh much less than Wh = residual oil

The character ratio was selected in order to interpret zones where high methane values indicate a

less-dense fluid (using the Wh and Bh ratios) than actually exists in the reservoir.

Again

according to *Haworth, et al*, Ch may be interpreted as follows:

Ch < 0.5; the Wh and Bh interpretation is correct

Ch > 0.5; gas character indicated by Wh and Bh actually

Ch is only used when Wh and Bh indicate gas.

GAS DATA INTERPRETATION

Wetness $W_h = \frac{C_2 + C_3 + C_4 + C_5}{C_1 + C_2 + C_3 + C_4 + C_5} \times 100$

Balance $B_h = \frac{C_1 + C_2}{C_3 + C_4 + C_5}$

Character $Ch = \frac{C_4 + C_5}{C_3}$

Wh < 0.5 = very dry gas

Wh from 17.5 to 40 = oil

Wh from 0.5 to 17.5 = gas

Wh > 40 = residual oil

3. Continuous Depth plots : LM, LH, HM

Suggested Hydrocarbons Indication Formulae

Continuous plot of combinations of the heavier & lighter hydrocarbons at the Well site.

Continuous Plots for Hydrocarbon Detection:

Plot: 1

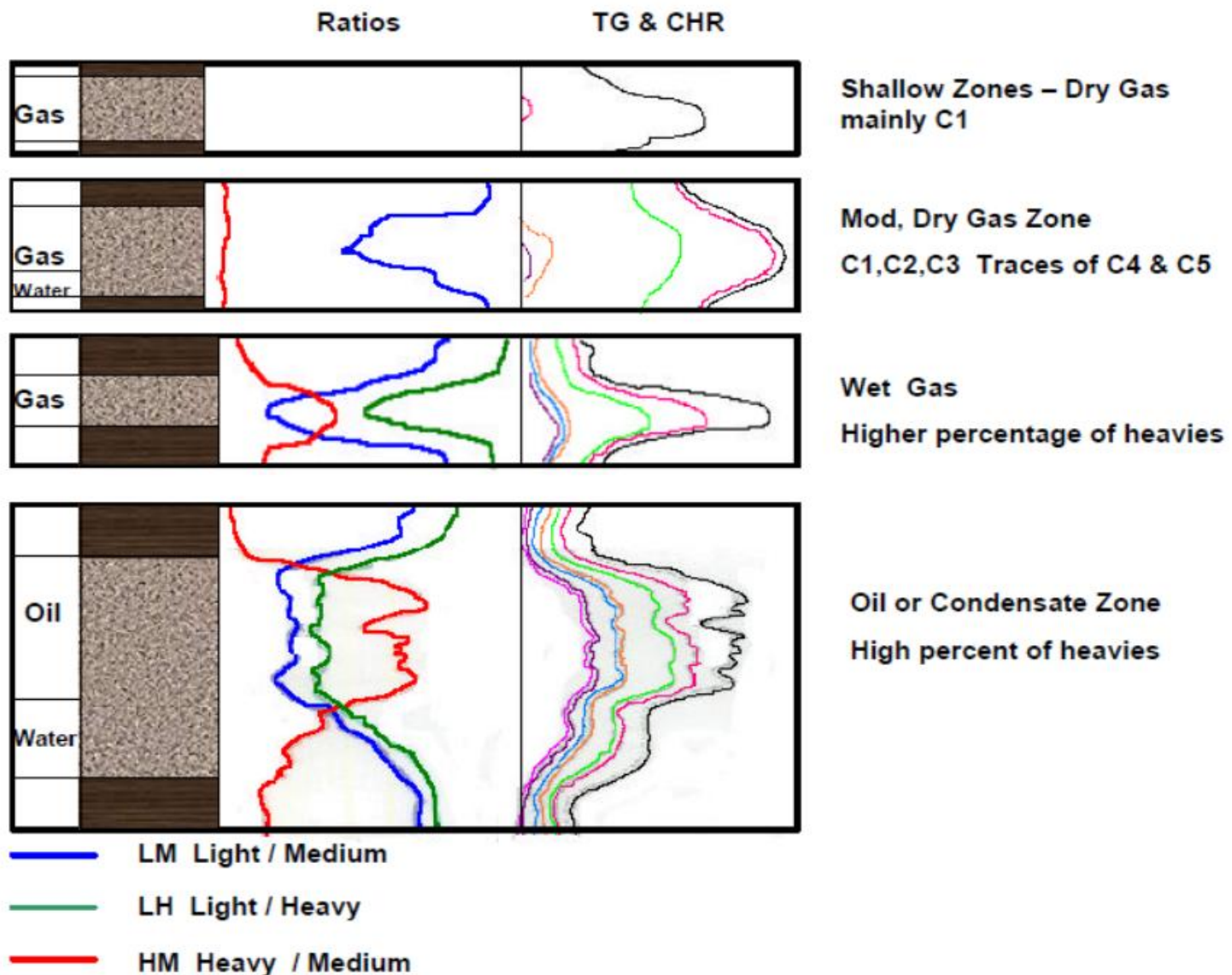
- | | | |
|------|-------------------------------------|-----------------|
| • Lh | $100 \times (C1+C2) / ((C4+ C5)^3)$ | Light to Heavy |
| • Lm | $10 \times (C1) / ((C2+ C3)^2)$ | Light to Medium |
| • Hm | $((C4+ C5)^2) / (C3)$ | Heavy to Medium |

Plot 2:

- | | | |
|-------|------------------------------|------------------------|
| • HCI | $(TG - (C1+C2)^{0.5}) / 500$ | Hydrocarbon Indicator. |
|-------|------------------------------|------------------------|

Values of TG, C1 C5 in ppm.

GAS DATA INTERPRETATION



Gas Data Interpretation

Fluid Indication

Dry gas zone : will contain only methane or very small quantities of heavier gases if at all. This type of gas zones are common in shallow depths, high temperature areas, immature sediments etc. No ratio curve will show any deflection. HM will remain near zero. LM and LH will be very high. Total gas will be high.

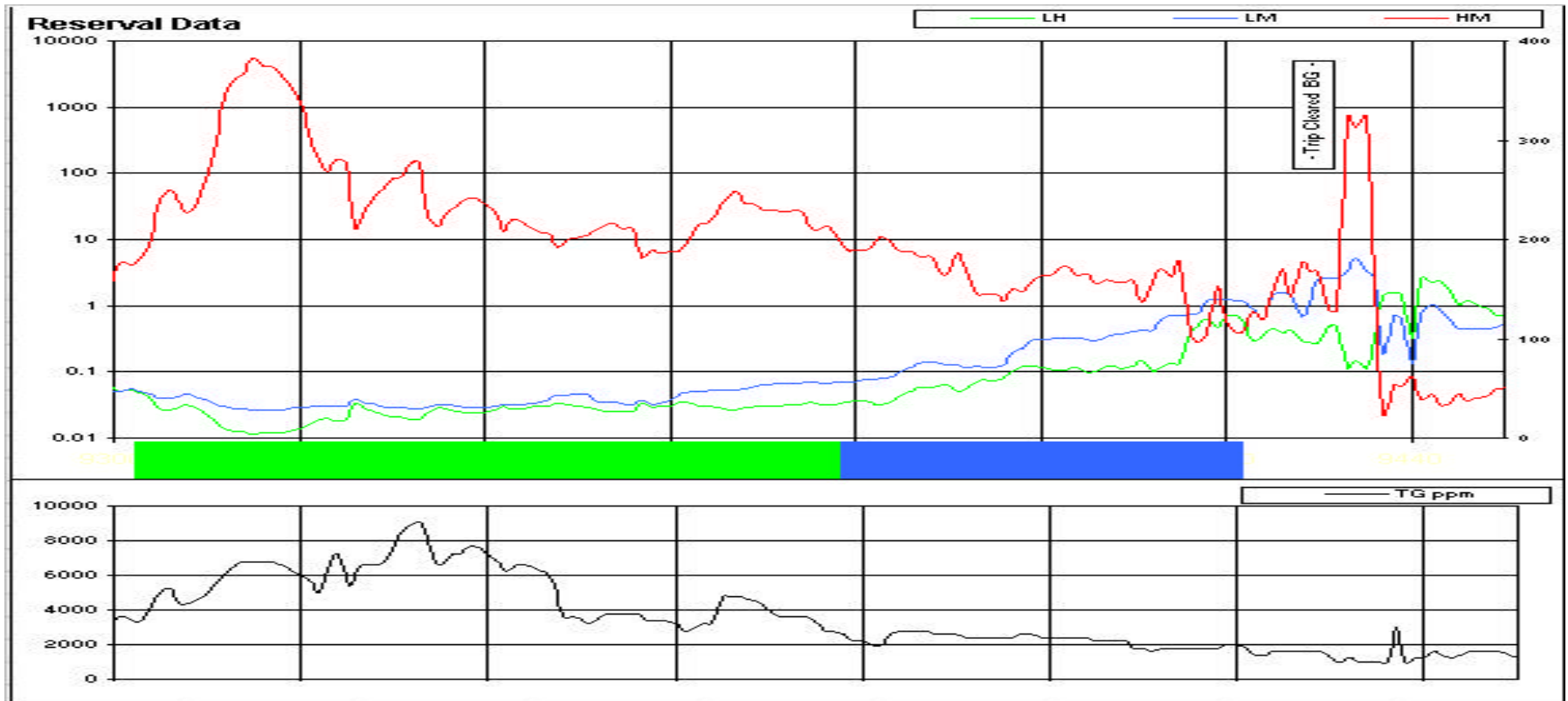
Moderate Gas Zone : This type of zones show low deflections of HM and low decrease in LH. LM decrease is pronounced and can be easily picked up. Gas water contacts can be made out.

GAS DATA INTERPRETATION

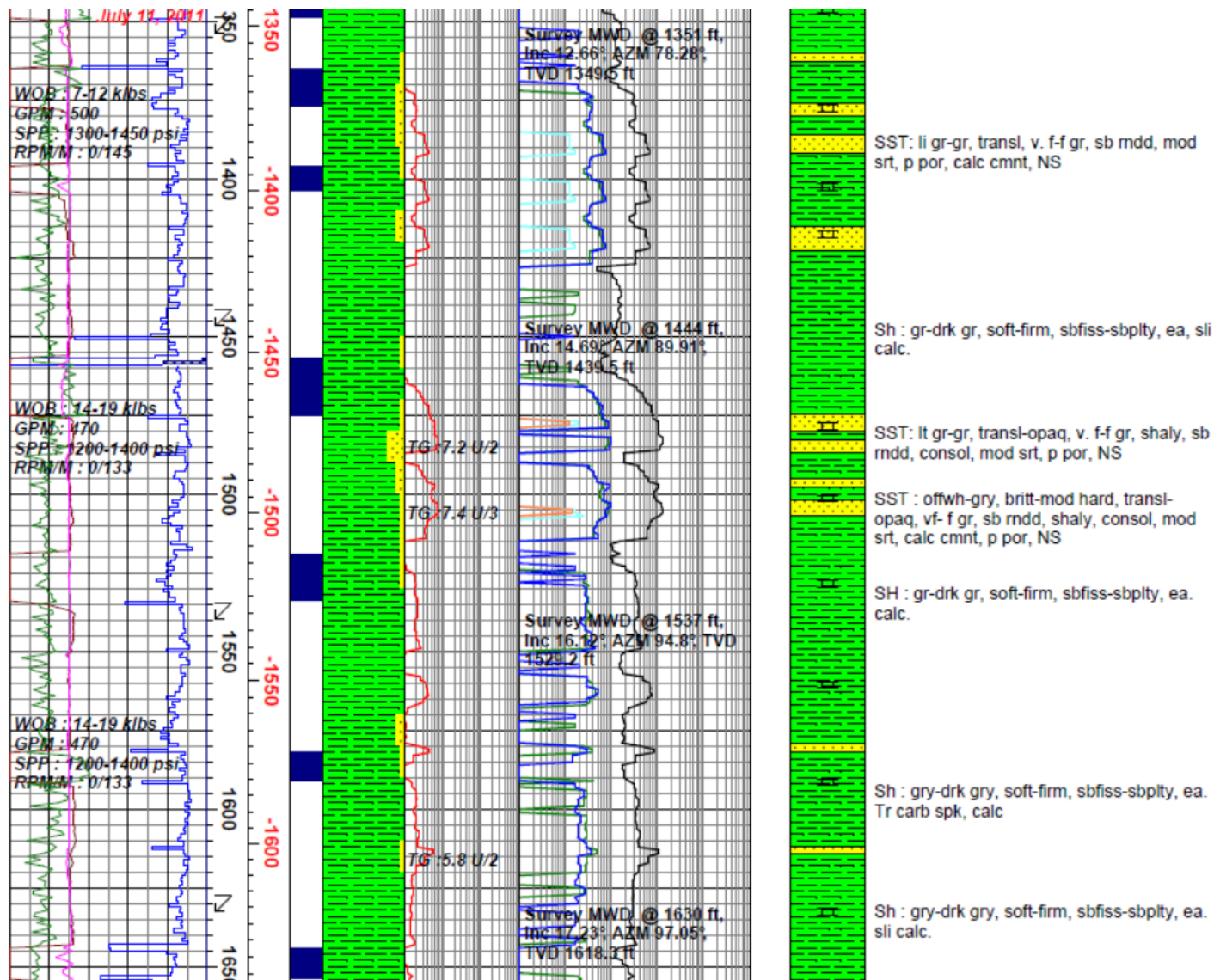
Oil zones – condensate to low API crude.

HM LH and LM all show strong deflections. Total gas or CTG will show moderate peak, compared to gas zone containing some oil.

The Oil water contact can also be made out from the deflections and TG which comes down.



GAS DATA INTERPRETATION



GAS DATA INTERPRETATION

